Securing Large Language Models: A Survey of Watermarking and Fingerprinting Techniques

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Abstract

This survey provides a comprehensive, critical analysis of watermarking and fingerprinting techniques for securing large language models (LLMs), addressing urgent challenges in provenance tracking, copyright enforcement, model ownership verification, and regulatory compliance. We rigorously distinguish *text watermarking*—imperceptible, statistically verifiable signal embedding into LLM-generated token sequences—and *model fingerprinting*—tamper-resistant identifier binding to model weights, behavior, or training dynamics—establishing a precise conceptual boundary essential for evaluation and governance. Through a novel three-axis taxonomy (Target, Embedding Mechanism, Verification Paradigm), we unify 32 state-of-the-art methods, exposing systematic, non-compensatory trade-offs among detectability, fidelity, robustness (against paraphrasing, translation, fine-tuning, and LLM-powered attacks like SIRA and GRI), computational overhead, and deployability. Empirical analysis reveals foundational limits: Easymark's impossibility theorem proves erasure resistance cannot coexist with both semantic fidelity and statistical detectability under adversarial distortion, while Water-Probe demonstrates that high detectability compromises LLM imperceptibility. We introduce a standardized five-dimensional evaluation framework, benchmarking gaps including the absence of merged-model testbeds and inconsistent cross-lingual protocols, and identify incentive misalignment—among providers, platforms, users, and regulators—as the primary barrier to real-world adoption. Finally, we chart a forward-looking agenda centered on provable robustness, multimodal generalization (e.g., VLA-Mark, code watermarking), adaptive threat resilience, ISO/IEC JTC 1/SC 42–aligned standardization, and governance-ready infrastructure (e.g., FedZKP, TokenizedModel, Proof-of-Training). This work establishes watermarking and fingerprinting as indispensable, co-designed pillars for trustworthy, accountable, and regulation-compliant LLM ecosystems.

1 1. Introduction

1.1 The LLM Security Imperative

Large language models (LLMs) introduce systemic security vulnerabilities rooted in their statistical nature and opaque generation mechanisms: hallucination, embedded societal bias, unreliable synthetic text detection, and susceptibility to adversarial manipulations—including jailbreaking, prompt injection, and API distillation [1, 2]. Unsecured deployment enables disinformation ecosystems where watermarked English outputs are translated into Chinese via round-trip translation (RTT), obfuscating provenance and undermining authenticity verification under regulatory regimes such as the EU AI Act and NIST AI RMF [3]. Academic integrity erodes as students and researchers deploy high-fidelity machine-generated text without attribution [4], while commercial intellectual property is compromised through unauthorized training on copyrighted articles and source code—and increasingly evasive paraphrasing attacks that bypass conventional detectors [5]. Provenance collapse intensifies in multilingual contexts: RTT-based text laundering degrades traceability for low-resource

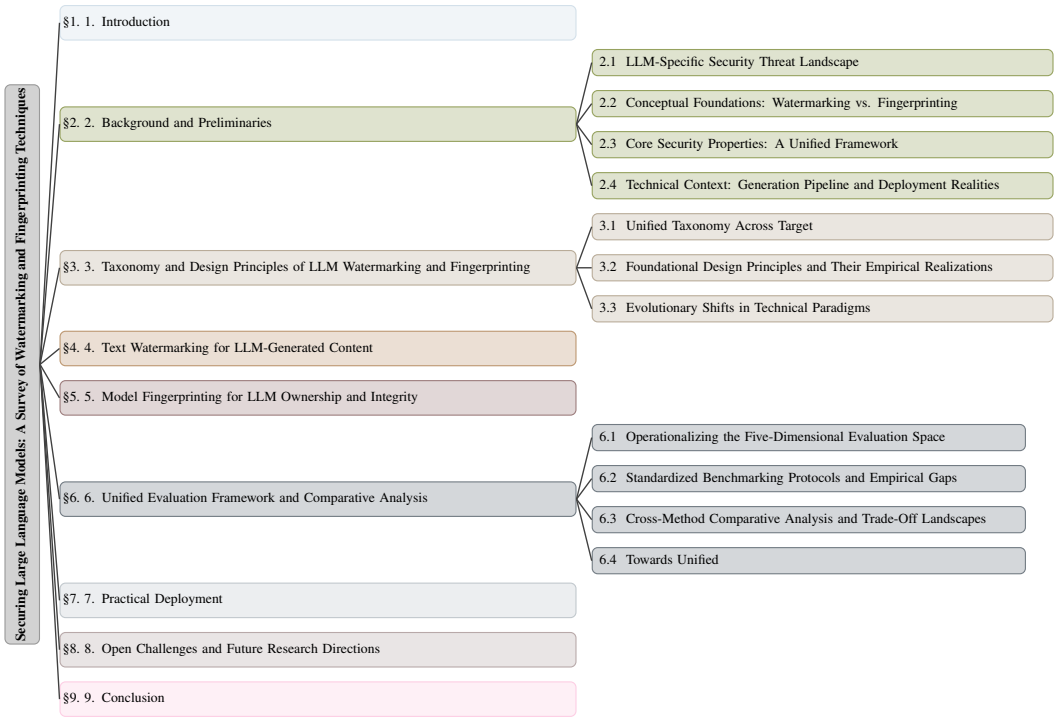


Figure 1: chapter structure

languages like Bangla, weakening evidentiary accountability in journalism and public discourse [6]. Personalized and domain-specific LLMs further compound these risks by introducing novel tensions among content authenticity, IP enforcement, and regulatory compliance for AI-generated outputs [7]. Underlying all these threats is the structural absence of intrinsic provenance markers in digital text—the primary carrier of information—enabling unattributed copying across platforms [4]. Although watermarking techniques demonstrate strong technical performance in detectability, robustness, and fidelity [8, 5], real-world adoption remains negligible due to misaligned stakeholder incentives, covert deployment practices that expose statistical biases under repeated sampling [9], and the lack of standardized, auditable verification protocols—leaving copyright protection, dataset provenance, and regulatory auditability fundamentally unenforceable.

1.2 Dual-Scope Clarification and Conceptual Boundaries

Text watermarking and model fingerprinting represent logically disjoint security primitives, rigorously distinguished by target locus, operational mechanism, and normative purpose. Text watermarking embeds statistically verifiable, imperceptible signals *exclusively into token sequences generated by LLMs*, enabling downstream attribution, copyright enforcement, and provenance tracking—without altering model parameters, architecture, or inference behavior [10]. Output-level embedding supports multi-bit payloads (e.g., model identifiers, timestamps, user hashes) via logits perturbation (CTWL [11]), vocabulary permutation (WALL [5]), syntactic filtering (PersonaMark [7]), or context-aware lexical substitution (CASIS [12]). In contrast, model fingerprinting embeds tamper-resistant identifiers *within the model itself*—either as parameter-space perturbations (PLMmark [2]), behaviorally encoded statistical deviations (SAC [2]), or training-dynamics commitments (PoT [2])—to verify ownership, detect unauthorized replication or fine-tuning, and ensure integrity across deployment variants. ImF achieves verification solely through behavioral signatures without internal model access, preserving structural integrity while avoiding output-level signal injection [13]; DAWN embeds cryptographically secured backdoor watermarks in surrogate DNN models by inducing deterministic mispredictions on a small trigger set, enabling ownership proof without modifying weights or text generation logic [10]; TokenizedModel formalizes this boundary by embedding watermarks either in weights or inference behavior—via feature regularization or backdoor triggers—and verifying provenance without altering outputs [14]. Steganography, though superficially similar in its use of lexical substitution, operates under a secrecy-oriented threat model and covert communication

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objective—rendering it conceptually orthogonal to both attribution-focused text watermarking and integrity-focused model fingerprinting; it is retained only for boundary clarification and excluded from core technical analysis [12].

1.3 The Multidimensional Trade-Off Landscape

No existing watermarking or fingerprinting technique simultaneously optimizes detectability, fidelity, robustness, computational efficiency, and black-box deployability—each advance in one dimension incurs measurable degradation in at least one other, constrained by the limited space of semantic-preserving modifications in natural language and model behavior [15]. Water-Probe reveals a triadic tension among watermark detectability, text imperceptibility (fluency/coherence preservation), and LLM imperceptibility (resistance to black-box model identification via repeated sampling), showing that statistical distinguishability can inadvertently leak model identity without adaptive strength modulation [9]. This constraint is formalized in Easymark's Theorem 3.4—the Impossibility Theorem for meaning-preserving watermarks—which proves erasure resistance cannot be guaranteed without sacrificing either semantic fidelity or detectability under adversarial semantic distortion [8]. Empirically, PersonaMark achieves high detectability (F1 = 0.942, AUC = 0.987) and preserves fidelity (perplexity = 2.74 vs. KGW's 12.32; consistent sentiment polarity and Flesch Reading Ease) and robustness against synonym substitution up to 40%, yet explicitly sacrifices payload capacity and cross-lingual generalizability to prioritize behavioral coherence [7]. WALL attains state-of-the-art scalability (>$10^{130,000}$ unique IDs), near-perfect verifiability (AUROC 1.0 post-fine-tuning), and sub-40ms CPU verification latency—but only by restricting embedding to high-probability token regions, limiting resilience to aggressive paraphrasing and RTT attacks that redistribute probability mass across low-entropy alternatives [5]. X-Mark improves detectability (AUROC 0.9124–0.9886) while reducing task accuracy degradation on GSM8K from 50.0% to 18.3% and TruthfulQA truth×info scores from 55.9% to 43.7%, yet does so by relaxing strict token-level constraints in favor of instruction-aligned latent-space deviations—diminishing resistance to fine-grained adversarial removal [16]. Cross-lingual consistency introduces a new axis: X-SIR enhances robustness under cross-lingual RTT attacks (CWRA), yielding +0.221 AUC gain without fluency or task accuracy loss—but requires explicit semantic clustering of multilingual vocabulary, increasing detector memory footprint and reducing compatibility with lightweight, quantized inference engines [3]. Layered Watermarking (LWM) lifts post-RTT detection accuracy to 40–50% (a 3–4× improvement over KGW/EXP) while introducing only moderate semantic degradation (ROUGE-L < 0.04, perplexity < 0.8), yet demands coordinated integration across sampling, lexical, and syntactic layers—compromising deployability in API-constrained environments mandating modular, single-point interventions [6]. Crucially, technical robustness alone is insufficient: incentive misalignment among providers, platforms, users, and regulators imposes systemic deployment constraints that render even theoretically optimal trade-off configurations non-viable without co-designed governance mechanisms and stakeholder-aligned enablers [17].

1.4 Structure of the Survey

The survey provides a rigorous, layered exposition of watermarking and fingerprinting for LLMs—from foundational abstractions to sociotechnical implementation—organized into eight logically progressive sections. Section 2 establishes shared technical and conceptual preliminaries: formally defining LLM-specific threat vectors (e.g., API distillation, unauthorized fine-tuning on proprietary data [2], provenance obfuscation in collaborative editing [18]), clarifying the formal distinction between text watermarking (statistically verifiable signal injection into token sequences [10]) and model fingerprinting (tamper-resistant identifier binding to weights, behavior, or training dynamics), and unifying the five core security properties—imperceptibility, robustness, detectability, fidelity, and verifiability—within a coherent measurement framework. Section 3 introduces a novel three-axis taxonomy organizing both domains along Target (text output vs. model weights/behavior), Embedding Mechanism (sampling-based, vocabulary partitioning, steganographic substitution, contrastive deviation, instruction-triggered, CoT-aligned), and Verification Paradigm (key-dependent/keyless, black-box/white-box, training-free/model-modifying), then distills empirically grounded design principles—including entropy-aware token weighting (KGW [8], EXP [3]), probability-balanced vocabulary partitioning (SynthID [15]), semantic coherence as robustness anchor (ImF [13]), adversarial alignment in threat modeling (SAC [2]), and statistical independence of watermark bits (CTWL [11]). Sections 4 and 5 constitute the technical core: Section 4 surveys text watermarking methods

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by embedding mechanism—sampling-based (KGW, SynthID-Text, EXP), payload-rich codable schemes (CTWL, Balance-Marking), deep learning–driven approaches (DeepTextMark, EWD [5]), and linguistic/steganographic techniques (Tracing Provenance via LS, BanglaLorica)—with critical analysis of cross-lingual failure modes and vulnerability to detection-aware probing [9]. Section 5 comprehensively reviews model fingerprinting across embedding loci: parameter-space perturbations (PLMmark, Quantization Watermarking), behavior-based methods without parameter modification (SAC, MetaV, Holmes [2]), implicit semantic signatures (ImF, Explanation-as-Watermark [13]), and provenance-centric governance frameworks (PoT, FedSOV [2]). Section 6 proposes and applies a standardized five-dimensional evaluation framework—operationalizing detectability (AUC, precision/recall), fidelity (perplexity, ROUGE-L, human-rated fluency), robustness (success rate under paraphrasing, translation, fine-tuning, merging, SIRA/GRI), computational overhead (latency delta, detector memory), and deployability (API compatibility, language support)—to expose systematic trade-offs, benchmarking gaps (e.g., absence of merged-model testbeds [5], inconsistent attack budgets [17]), and cross-method performance landscapes (e.g., SynthID-Text's high detectability versus PersonaMark's cross-lingual fragility). Section 7 shifts focus to real-world viability, analyzing stakeholder incentive misalignment (providers' fear of watermark leakage via Water-Probe [9], platform resistance to mandatory detection tooling [17]), sociotechnical deployability constraints (legal uncertainty in copyright litigation [15], ethical concerns about covert fingerprinting [14]), and incentive-aligned enablers (Water-Bag's per-user key randomization [9], X-SIR's cross-lingual robustness [3], layered watermarking in BanglaLorica [6]). Finally, Section 8 identifies foundational open challenges—including theoretical limits (Easymark's impossibility theorem [8]), multimodal generalization (code, vision-language, audio), adaptive threat resilience against LLM-powered paraphrasers and gradient-based removal (GRI-2 [2]), standardization pathways (ISO/IEC JTC 1/SC 42 profiles [17]), and governance-ready infrastructure (FedZKP, Tokenized Model, Watermark-as-a-Service). The survey concludes by synthesizing contributions toward unified taxonomy, evidence-based trade-off characterization, and a forward-looking agenda centered on provable robustness, interoperable standards, and sociotechnical accountability.The following sections are organized as shown in Figure 1.

2 2. Background and Preliminaries

2.1 LLM-Specific Security Threat Landscape

Large language models introduce security threats intrinsic to their statistical generation, emergent scale-dependent behaviors, and deployment across high-stakes sociotechnical domains—not merely conventional software vulnerabilities [1]. Hallucinated factual claims, deepfake text indistinguishable from human-authored content, and AI-amplified phishing or social engineering campaigns exploit LLMs' capacity for coherent, context-aware, and stylistically adaptive output [1]. In software development, AI-generated code exhibits severe provenance obfuscation: adversaries embed watermarked or fingerprinted LLM outputs in forensic tooling, malware analysis pipelines, and academic submissions to evade attribution and undermine integrity verification where source accountability is critical [19]. Model theft via black-box API distillation—e.g., KnockOff or PRADA-style query-based extraction—enables high-fidelity surrogate model replication without weight access, directly compromising proprietary intellectual property with minimal computational investment [2]. Teacher model exposure during transfer learning further enables adversaries to track proprietary teachers through student behavior, inherit latent vulnerabilities (e.g., backdoors), and facilitate downstream attacks including model stealing and adversarial example generation [20]. Unauthorized fine-tuning on proprietary data introduces a dual-layered threat: adversaries fine-tune black-box LLM services using watermarked outputs as training inputs to extract latent knowledge or replicate functionality while evading detection—WALL explicitly addresses this vector by embedding identifiers that persist through such fine-tuning cycles [5]. Provenance obfuscation intensifies in multilingual collaborative editing, where round-trip translation (RTT) attacks—e.g., Bangla $\rightarrow$ English $\rightarrow$ Bangla—degrade watermark integrity while preserving semantic fidelity and readability, particularly for morphologically rich, non-Latin-script languages; similarly, English-generated watermarked text translated into Chinese, Japanese, French, or German evades detection under regulatory scrutiny [3, 6]. These cross-lingual watermark removal attacks (CWRA) exploit structural asymmetries in vocabulary alignment and tokenization, rendering many state-of-the-art watermarks ineffective outside monolingual English contexts [3]. Regulatory compliance risks stem directly from unattributed synthetic content: frameworks including the EU AI Act and NIST AI RMF mandate traceability and accountability for AI-generated

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outputs, yet current deployments lack standardized, auditable provenance signals—exposing organizations to legal liability, reputational damage, and enforcement actions when synthetic text enters journalism, academia, or governance without verifiable origin [1, 3]. Critically, these threats are systemically intensified by incentive misalignment: providers avoid voluntary watermark deployment due to fears of black-box watermark identification—where adversaries infer presence, type, or even key solely from input–output behavior—thereby exposing competitive differentiators and enabling targeted evasion [9]. Moreover, existing black-box fingerprinting schemes claim robustness only against benign perturbations (e.g., prompt-wrapping, light fine-tuning, model merging), not against systematic adversarial attacks designed to erase behavioral signatures—leaving ownership verification vulnerable to sophisticated removal techniques like SIRA, which performs black-box paraphrasing without access to the watermark algorithm, secret key, or detector implementation [21]. These interlocking vectors define an LLM-specific threat landscape in which technical security properties must be co-designed with operational realism, adversarial awareness, and regulatory enforceability.

2.2 Conceptual Foundations: Watermarking vs. Fingerprinting

Text watermarking is formally defined as statistically verifiable, imperceptible signal injection into token sequences generated by large language models, where encoding $Enc(x_{prompt}, m)$ maps a prompt and message $m \in \mathcal{M}$ to a watermarked sequence t, and decoding $Dec(t)$ recovers m with quantifiable confidence—generalizing beyond binary payloads to arbitrary finite message spaces (e.g., $|\mathcal{M}| = 2^{20}$) [11]. This definition excludes modification to model weights, inference logic, or training dynamics, anchoring the mechanism strictly to the output generation pipeline [10]. SynthID-Text instantiates this via generative sampling modification—perturbing logits to bias selection toward statistically distinguishable "green" tokens—while preserving fluency and task fidelity without embedding identifiers in the model itself [22]. PersonaMark implements it through sentence-level structural hashing of dependency-parsed candidates, filtering outputs whose syntactic representation hashes to '1' under a user-ID-augmented cryptographic function—requiring no logits perturbation or parameter alteration [7]. WALL achieves payload-rich encoding via deterministic, hash-driven vocabulary permutations and orthogonal logit-space perturbations, enabling detection robust to surface-form transformations while remaining confined to the decoding stage [5]. Easymark satisfies the formalism through Unicode codepoint substitution—preserving semantics, visual appearance, and statistical quality metrics (BLEU, perplexity)—demonstrating that imperceptibility extends beyond linguistic coherence to typographic and orthographic fidelity [8]. X-Mark embeds signals via green-list biasing with detection grounded in z-statistic rejection of the null hypothesis (H_0: no watermark), satisfying both statistical verifiability and imperceptibility requirements under adversarial scrutiny [16]. CASIS realizes text watermarking as meaning-preserving lexical substitution—guaranteeing imperceptibility by maintaining syntactic/semantic structure and character count—while supporting verifiability without parameter modification [12]. DL-TW operates at the sentence level using embedding-guided synonym substitution and transformer-based blind detection—requiring neither original text nor internal model states [18]. DeepTextMark extends the formalism by embedding via semantically and grammatically congruent synonym replacements, with verification delegated to a fine-tuned BERT classifier operating solely on the watermarked text—eliminating dependence on secret keys, green/red token sets, or model internals [23]. PNW formalizes an end-to-end black-box variant wherein neural generation and detection co-adapt without explicit token labeling or shared secrets, demonstrating that statistical verifiability need not rely on predefined token partitions [24]. VLA-Mark generalizes the definition to multimodal settings, embedding statistically verifiable, imperceptible signals into token sequences of vision-language aligned multimodal large models (VLAMMs), affirming that text watermarking remains fundamentally output-centric even in cross-modal architectures [25]. In contrast, model fingerprinting is defined as the tamper-resistant, behaviorally encoded embedding of identifiers *within the model itself*—either as parameter-space perturbations (e.g., PLMmark's contrastive weight modulation bound to RSA digital signatures and SHA256 hash chains, mapping owner identity to vocabulary tokens and feature-space geometry rather than output labels [26]), inference-time statistical deviations induced by trigger sets (e.g., SAC's adversarially constructed inputs eliciting distributional shifts in logits [2]), or training-dynamics commitments (e.g., PoT's cryptographic trajectory logging [2])—with the normative objective of verifying model ownership, detecting unauthorized replication or fine-tuning, and ensuring integrity across deployment variants [2]. Explanation-as-Watermark (EaaW) embeds a k-bit watermark in the sign of the feature attribution explanation of a secret trigger sample, constituting a behaviorally encoded identifier without altering model predictions [27]. FedSOV embeds fingerprints in model weights via regularization-based

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watermarking of BatchNorm scale parameters, requiring both successful extraction (within near-collision tolerance) and cryptographic signature validation to verify ownership—distinguishing it from text watermarking by targeting model provenance rather than output traceability [28]. TokenizedModel formalizes model fingerprinting through two complementary loci: feature-based watermarks embedded via regularization term $\lambda||AW - K||^2$ in the loss function, and backdoor-based watermarks embedded through poisoned dataset construction $\alpha D + (1 - \alpha)T$ with target labels—both anchored to on-chain metadata without altering output token sequences [14]. MOVE embeds external features into training data to induce unique behavioral signatures detectable via meta-classifier on gradients (white-box) or prediction-difference vectors (black-box), exemplifying behaviorally encoded identification rooted in training dynamics [29]. Holmes verifies ownership by detecting inherent dataset-specific features through output-behavior analysis—embedding no identifier in parameters or training process, achieving verification without model modification, and explicitly avoiding watermarking's structural interventions [30]. Behavior-based fingerprinting without parameter modification is further instantiated by adversarial examples serving as fingerprints [31], reinforcing that behavioral encoding suffices for tamper-resistant identification. FedZKP implements model fingerprinting—not text watermarking—by embedding a tamper-resistant, behaviorally encoded identifier (hash $h = H(A||y)$ of an xLPN problem instance) into BatchNorm scaling parameters, with verification relying on zero-knowledge proof of the secret xLPN solution (s, e), thereby achieving model provenance rather than output traceability [32]. Proof-of-Training (PoT) is formally distinguished from model fingerprinting: while fingerprinting embeds tamper-resistant identifiers in weights or behavior for ownership/integrity, PoT verifies authenticity of the *training process itself* (data, algorithms, trajectory) by leveraging the computational cost asymmetry between honest training and forging—making it a complementary, higher-level provenance mechanism [33]. This tripartite conceptual architecture—text watermarking (signal in outputs), model fingerprinting (identifier in model), and PoT (process-level certification)—establishes a rigorous, operationally grounded boundary essential for precise threat modeling, evaluation design, regulatory interpretation, and sociotechnical governance.

2.3 Core Security Properties: A Unified Framework

Imperceptibility, robustness, detectability, fidelity, and verifiability constitute five essential, interdependent security properties that jointly determine the operational viability of watermarking and fingerprinting techniques for large language models—each formally grounded in reproducible measurement protocols and empirically constrained by fundamental trade-offs. Imperceptibility is violated not only when human observers detect artifacts but also when LLMs themselves reliably identify watermarks via native pattern recognition mechanisms; [34] demonstrates that Unicode-level statistical anomalies—such as non-uniform codepoint distributions in Easymark's substitution space—can be exploited by LLMs to infer watermark presence, thereby redefining imperceptibility as a dual-domain requirement encompassing both human and model-native perception. This extends beyond linguistic fluency to typographic fidelity (Easymark preserves zero BLEU/perplexity change and visual indistinguishability [8]), semantic continuity (cosine similarity > 0.97 in DL-TW [18], Sentence Meaning Similarity 0.9765–0.9892 in DeepTextMark [23]), and syntactic coherence (PersonaMark maintains Flesch Reading Ease and dependency-structure consistency [7]). CASIS achieves imperceptibility through semantic-relatedness–filtered substitutions (SR > 0.95) validated by human evaluation (4.3+ mean rating), while DAWN satisfies it via 0.5% utility degradation and indistinguishability under input perturbations [10]. VLA-Mark preserves text-visual coherence and fluency (7.4% lower perplexity vs. baselines) [25], and Unicode homoglyph substitution attains visual indistinguishability across 10 major web/desktop fonts with zero character-length overhead [4]. Crucially, imperceptibility must also hold against behavioral inference: Holmes achieves imperceptibility with no human- or task-visible artifacts while preserving full victim model accuracy and task performance [30]; MOVE ensures imperceptibility by maintaining near-identical victim model accuracy (91.79% vs. 91.99% on CIFAR-10) without fluency or coherence impact [29]; FedZKP guarantees imperceptibility through absence of human-detectable artifacts [32]; and TMFA enforces imperceptibility by restricting paragraph break relocations exclusively to positions with 20–80% native-Russian-speaker insertion rates—ensuring changes are indistinguishable from natural editorial variation without semantic, syntactic, or lexical distortion [35]. Robustness is defined as survival under semantic-preserving edits—including synonym substitution (up to 40% in PersonaMark [7]), fine-tuning, merging, and LLM-powered paraphrasing—but must be explicitly decomposed across orthogonal threat surfaces: cross-lingual consistency constitutes a distinct robustness property requir-

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ing signal preservation under machine translation, which cannot be subsumed under monolingual paraphrasing robustness and is systematically violated by embedding-time methods like KGW and EXP in low-resource languages such as Bangla, where RTT collapses detection accuracy below 20% due to lexical substitution, syntactic reordering, and morphological variation [6]. Empirical quantification includes Mean Impact of Attack (mIOA) in DL-TW [18], AUROC 1.0 post-fine-tuning in WALL [5], and resilience to SIRA and GRI adversarial removal attacks [21, 13]; ImF further achieves robustness against Generation Revision Intervention (GRI) and model merging while improving TruthfulQA accuracy [13]. Critically, robustness must be defined to include survival under white-box adversarial settings where an adversary aims to evade verification while preserving utility; existing schemes fail this requirement due to fundamental vulnerabilities in verbatim response matching, output overconfidence, unnatural fingerprint queries, and statistical signature leakage [36]. SIRA directly challenges the core security property of robustness—demonstrating that state-of-the-art watermarks (KGW, SIR, EXP, EWD, etc.) fail under a targeted, self-information-guided paraphrasing attack while preserving semantic fidelity and text quality [21]. CASIS retains 0% bit error rate under print-camera, screen-camera, OCR, and retyping attacks [12], DAWN survives pruning, noise injection, adversarial fine-tuning, and distributed collusion [10], and attribution accuracy remains >88% at 20-class under aggressive semantic-preserving transformations including minification, mangling, obfuscation, and deobfuscation [19]. TMFA exhibits robustness to fine-tuning while remaining vulnerable to input distortion defenses [20], and EaaW achieves Watermark Success Rate > 0.85 after fine-tuning, pruning, unlearning, and overwriting [27]. FedSOV formally grounds robustness in a security boundary $\varepsilon(n)$ derived from near-collision-resistant hashing (e.g., $\varepsilon(2048) = 82.1\%$) and achieves >99% watermark detection rate under fine-tuning, pruning, and targeted noise [28]; FedZKP attains 100% detection rate under fine-tuning and pruning [32]; Holmes resists false positives from distributionally similar benign models and survives soft distillation, fine-tuning, and data-free distillation [30]; and MOVE correctly identifies all tested model stealing types—including distillation, fine-tuning, zero-shot, label-query, and logit-query attacks [29]. Detectability demands statistical confidence in verification, operationalized through log-likelihood ratio tests (KGW [8]), z-statistic hypothesis rejection (X-Mark [16]), or binary classification accuracy (78.83% in DL-TW [18], >90% AUC in DeepTextMark [23]); critically, it requires quantification of minimum token requirements for reliable detection—a metric underexplored in current literature but essential for practical deployment [15]. EWD improves detectability specifically under low-entropy conditions via entropy-proportional weighting that maximizes signal-to-noise ratio without increasing Type-I error or degrading fidelity [37]. CASIS enables deterministic synchronicity-guided extraction with zero-bit-error-rate [12], DAWN achieves ownership verification with false positive probability < 2^{-64} [10], and EaaW attains statistical significance via chi-squared test on extracted watermark bits (p-value < 0.01) [27]. VLA-Mark achieves 98.8%+ AUC across four VLAMMs and 99.7% AUROC variant [25], while SAC-m achieves AUC-ROC score up to 1.0 [38]. Holmes achieves p-values as low as 10^{-20} under Data Distill on ImageNet with $\alpha = 0.01$ [30], and MOVE attains statistical significance with p-value < 0.01 under one-tailed paired T-test [29]. Fidelity captures functional integrity preservation across multiple axes: perplexity (no increase >0.3 in Balance-Marking [11], < 0.8 in LWM [6]), task accuracy (GSM8K degradation reduced from 50.0% to 18.3% in X-Mark [16]), mathematical reasoning and safety refusal functionality (quantified jointly via Verification Success Rate (VSR) and task accuracy under model merging [39]), grammatical integrity (POS/number-aware synonym selection in DeepTextMark [23]), and fluency (preserved in ImF despite behavioral watermarking [13]). DAWN imposes zero modification to the student model yet incurs only 0.03–0.5% victim model accuracy drop [10], EaaW introduces <1% accuracy drop in CV and minimal PPL increase in NLP [27], and VLA-Mark maintains semantic alignment (STS, BertScore) and task accuracy [25]. FedSOV preserves fidelity with <2% main-task accuracy drop [28], FedZKP incurs <2% main-task accuracy drop [32], TM-PoT ensures high mutual information between data and trajectory to support accurate retraining [33], and Holmes preserves full victim model accuracy and task performance [30]. Verifiability encompasses three orthogonal dimensions: key dependency (keyless in PersonaMark [7], Balance-Marking [11], PNW [24], DeepTextMark [23], and Easymark [8] versus key-dependent in SynthID-Text [22] and WALL [5]), access mode (black-box compatible in all aforementioned methods—including training-free detection requiring only raw token IDs in PNW [24], sentence parsing in PersonaMark [7], and green-list token frequency in X-Mark [16]), and training requirements (training-free in Balance-Marking's proxy-LM decoupling [11], WALL's public-key-agnostic verification [5], DL-TW's add-on deployment [18], and PLMmark's black-box verification using only final-layer predictions and owner's public RSA key [26]). CASIS enables keyless, black-box–compatible extraction requiring only the watermarked text and public BERT/RoBERTa models

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[12], DAWN supports public bulletin board registration and TEE-based verification [10], TMFA requires only top-1 labels [20], and EaaW enables black-box, training-free extraction requiring only API queries to compute explanations from masked inputs [27]. In-context watermarking (ICW) further enables third-party-controlled, black-box, training-free verification without modifying model weights or decoding logic—aligning with stakeholder-specific verifiability needs for IP owners seeking keyless/public verification [17]. TokenizedModel satisfies verifiability as a core security property: verification is black-box (no access to model weights required during detection), training-free (uses pre-embedded watermarks), and keyless for end users—since watermark keys (K or T) remain confidential and only detection outcomes (pass/fail) are published on-chain via committee consensus [14]. MetaV satisfies black-box verifiability requirements: verification requires only black-box query access to suspect models, while target model access is white-box only during fingerprint construction [40]. FedSOV supports public-key–based, black-box verification requiring only model inference and challenge-response interaction [28], FedZKP achieves black-box, training-free, keyless verification (the xLPN public input (A_{agg}, y_{agg}) is publicly known) [32], and Holmes implements black-box, training-free, keyless verification via hypothesis testing on query outputs [30]. Dual-verification emerges as a new core security property: authentication requires simultaneous satisfaction of Label Accuracy (LA) > th_{LA} and Bit Error Rate (BER) < th_{BER}, enforcing both functional correctness and copyright bit recovery—extending beyond standard detectability and verifiability to include *fingerprint ownership verifiability* [31]. These properties are intrinsically coupled: Easymark's Theorem 3.4 proves that erasure resistance cannot be guaranteed without sacrificing either semantic fidelity or detectability under adversarial semantic distortion [8], while Water-Probe demonstrates that high detectability may compromise LLM imperceptibility [9]. Their unified operationalization—anchored in standardized metrics (AUC, perplexity delta, mIOA, STS, VSR, pass@10) and validated across diverse attack surfaces including cross-lingual RTT, model merging, and GRI—enables rigorous, comparable assessment across text watermarking and model fingerprinting domains.

2.4 Technical Context: Generation Pipeline and Deployment Realities

LLM decoding strategies—including temperature scaling, top-k, top-p (nucleus) sampling, beam search, and speculative decoding—impose non-negotiable architectural constraints on watermark and fingerprint design, as any intervention must preserve autoregressive token generation semantics while remaining compatible with hardware-accelerated inference stacks. Sampling-based watermarks such as KGW and EXP require precise control over logits at each decoding step to enforce green/red token set biasing, yet their statistical detectability degrades under high-temperature sampling or low-probability token selection unless entropy-adaptive thresholding is applied—as demonstrated by EWD's entropy-proportional weighting mechanism [37]. PersonaMark circumvents decoder-level fragility by operating at sentence boundaries via modified diversity beam search, avoiding logits perturbation entirely; however, its fidelity depends critically on dependency parser accuracy and punctuation-aware beam re-ranking, rendering it sensitive to aggressive lexical edits that disrupt syntactic structure [7]. WALL embeds signals autoregressively during token generation without altering model weights, enabling seamless integration into standard LLM inference APIs—whether deployed as closed commercial services (e.g., OpenAI API) or open-weight models (e.g., LLaMA-3, Phi-3)—and supporting quantized variants (e.g., AWQ, GPTQ) through its reliance solely on vocabulary-level logit-space operations rather than full-precision weight gradients [5]. DL-TW functions as a plug-in module that consumes raw token sequences from any standard decoding pipeline—sampling-based, deterministic, or nucleus-sampled—without requiring access to internal logits or hidden states, thereby achieving deployment agnosticism across both closed API and open-weight environments while maintaining sentence-level granularity [18]. Easymark operates entirely outside the decoding loop, performing Unicode codepoint substitution on finalized text outputs, which renders it invariant to all sampling parameters (temperature, top-k, nucleus p) and fully compatible with real-time streaming, client-side watermarking, and plain-text publishing workflows—including scenarios where only rendered HTML or PDF outputs are available [8]. In contrast, character-level methods like those in [4] decouple entirely from LLM internals, functioning as post-hoc text processors with zero dependency on decoding logic, thus supporting closed API, open-weight, and even static document publishing contexts without pipeline integration. For model fingerprinting, black-box deployment realities dominate: SAC requires only model outputs—logits, probabilities, or smooth-label distributions—and imposes no architectural assumptions, making it directly deployable in MLaaS settings where models are accessed exclusively via prediction APIs; its lightweight verification—requiring only 100–200 normal samples—achieves sub-5-second latency on CIFAR-10,

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satisfying strict operational constraints of production inference endpoints [38]. MetaV is explicitly engineered for such API-constrained environments, enabling black-box verification using only query-response pairs from suspect models, with fingerprint construction performed once in white-box mode and subsequent detection requiring no model access beyond standard inference calls [40]. Critically, quantization and adapter-based fine-tuning introduce structural erosion: parameter-space fingerprints like PLMmark must survive 4-bit AWQ quantization and LoRA adapter injection without signal collapse, while behavior-based fingerprints like Holmes and FedZKP are designed to persist under pruning, noise injection, and soft distillation precisely because they anchor identification to output distributions rather than weight precision [32, 30]. These deployment realities collectively enforce a hard boundary: watermarking techniques must either integrate natively into autoregressive decoding (WALL, PersonaMark), operate orthogonally to it (Easymark, [4]), or function as lightweight post-processing layers (DL-TW); meanwhile, model fingerprinting must guarantee verifiability under black-box API access, quantization, and adapter-based adaptation—or forfeit practical viability in real-world LLM ecosystems.

As illustrated in Figure 2, this hierarchical figure synthesizes the core conceptual architecture of modern LLM watermarking and fingerprinting, structured around three interlocking dimensions: (1) a unified three-axis taxonomy—distinguishing *target* (text, model, or prompt), *embedding mechanism* (sampling, partitioning, steganography, etc.), and *verification paradigm* (key dependency, access mode, training requirements); (2) five empirically grounded design principles—entropy-aware weighting, probability-balanced partitioning, semantic coherence, adversarial alignment, and bit independence—that collectively govern robustness and deployability; and (3) evolutionary paradigm shifts—from statistically conspicuous signals toward behaviorally fused, cross-lingually aware, operationally lightweight, and layered mechanisms—each catalyzed by real-world attack exposure (e.g., GRI, SIRA, CWRA) and regulatory demands (e.g., EU AI Act traceability). The structure reflects strict orthogonality, compositional synergy (e.g., WALL integrates three principles), and stakeholder-centered pragmatism, positioning the field at the frontier of provably robust, sociotechnically viable LLM provenance.

3 3. Taxonomy and Design Principles of LLM Watermarking and Fingerprinting

3.1 Unified Taxonomy Across Target, Mechanism, and Verification

A rigorous, orthogonal three-axis taxonomy resolves conceptual fragmentation in LLM watermarking and fingerprinting by decoupling *where* the signal is anchored, *how* it is instantiated, and *how authenticity is verified*. As illustrated in Figure **??**, this unified framework explicitly separates the **Target**, **Embedding Mechanism**, and **Verification Paradigm** axes—each comprising empirically grounded, mutually exclusive categories drawn from the state of the art. The **Target** axis distinguishes (i) *text output*, where signals reside exclusively in generated token sequences without weight or inference modification—exemplified by SynthID-Text [22], PersonaMark [7], Balance-Marking [11], WALL [5], DL-TW [18], DeepTextMark [23], PNW [24], Easymark [8], CASIS [12], VLA-Mark [25], and X-Mark [16]; (ii) *model weights or behavior*, where identifiers are bound to parameter-space perturbations (PLMmark [26]), trigger-induced statistical deviations (SAC [2], MetaV [40]), cryptographically grounded API-layer signatures (DAWN [10]), CoT-aligned behavioral optimization (ImF [13]), explanation-space modulation (EaaW [27]), dual shadow-model–induced output differences (Holmes [30]), external-feature-triggered imprinting (MOVE [29]), regularization-based embedding (FedSOV [28], TokenizedModel [14]), BatchNorm scaling perturbation (FedZKP [32]), or structural/semantic code signatures [19]; and (iii) *input prompt*, a newly formalized category instantiated by In-Context Watermarking (ICW), which embeds instruction-triggered signals via engineered user prompts—enabling third-party–controlled, zero-parameter-modification provenance with no decoding or model-internals intervention [17]. This tripartite target classification supersedes prior dual-axis frameworks that conflate granularity with locus [15] and explicitly accommodates prompt-level provenance control and code-structural attribution [19]. The **Embedding Mechanism** axis comprises six empirically grounded classes: (i) *sampling-based*, perturbing logits to bias autoregressive selection toward green tokens (KGW [8], EXP [3], SynthID-Text's tournament sampling [22]); (ii) *vocabulary partitioning*, enforcing probability-balanced green/red sets guided by proxy-LM likelihoods (Balance-Marking [11]) or cryptographic hashing over vocabulary indices (WALL [5]), extended by X-SIR's cross-lingual semantic clustering to satisfy RTT robustness [3];

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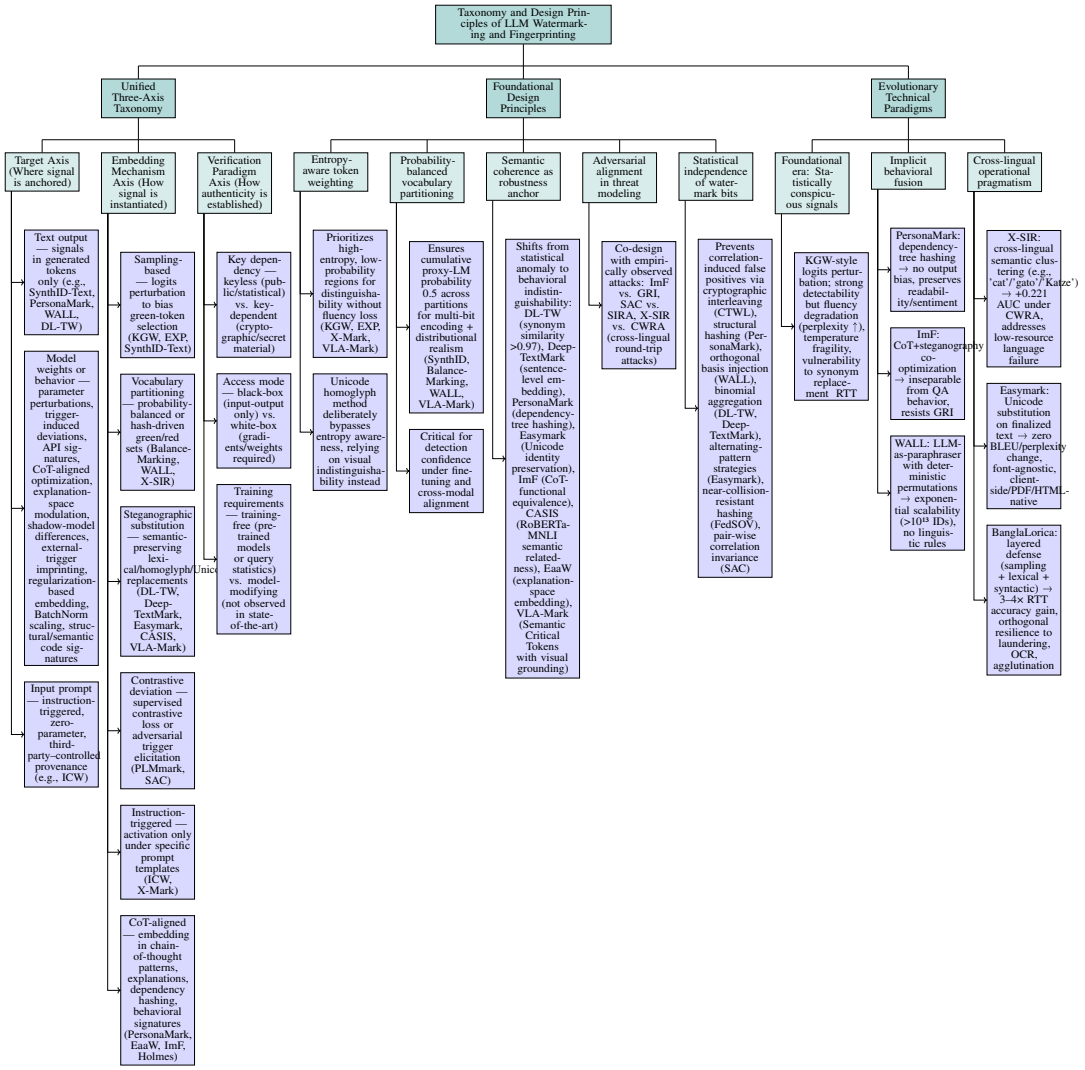


Figure 2: This hierarchical figure synthesizes the core conceptual architecture of modern LLM watermarking and fingerprinting, structured around three interlocking dimensions: (1) a unified three-axis taxonomy—distinguishing *target* (text, model, or prompt), *embedding mechanism* (sampling, partitioning, steganography, etc.), and *verification paradigm* (key dependency, access mode, training requirements); (2) five empirically grounded design principles—entropy-aware weighting, probability-balanced partitioning, semantic coherence, adversarial alignment, and bit independence—that collectively govern robustness and deployability; and (3) evolutionary paradigm shifts—from statistically conspicuous signals toward behaviorally fused, cross-lingually aware, operationally lightweight, and layered mechanisms—each catalyzed by real-world attack exposure (e.g., GRI, SIRA, CWRA) and regulatory demands (e.g., EU AI Act traceability). The structure reflects strict orthogonality, compositional synergy (e.g., WALL integrates three principles), and stakeholder-centered pragmatism, positioning the field at the frontier of provably robust, sociotechnically viable LLM provenance.

(iii) *steganographic substitution*, deploying semantic-preserving lexical, homoglyph, or Unicode-level replacements (DL-TW [18], DeepTextMark [23], Easymark [8], CASIS [12], VLA-Mark [25], Unicode homoglyph methods [4]); (iv) *contrastive deviation*, inducing behavioral signatures through supervised contrastive loss on weights (PLMmark [26]) or adversarial trigger-set elicitation of distribution shifts (SAC [2]); (v) *instruction-triggered*, activating watermarks only under specific prompt templates (ICW [17], X-Mark's instruction-conditioned green-list biasing [16]); and (vi) *CoT-aligned*, embedding identifiers in chain-of-thought reasoning patterns, feature attribution explanations (EaaW

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[27]), dependency-structure hashing (PersonaMark [7]), input-optimized behavioral signatures (ImF [13]), or dual shadow-model–induced output differences (Holmes [30]). This mechanism axis integrates threat-aware insights [2], extending beyond surface transformations to latent-space, behavioral, and instruction-conditioned embeddings—including AST, dataflow, and control-flow signatures for JavaScript attribution [19]. The **Verification Paradigm** axis structures authenticity establishment along three binary dimensions: (i) *key dependency*, distinguishing keyless schemes—relying solely on public models or statistical properties (PersonaMark [7], Balance-Marking [11], PNW [24], DeepTextMark [23], Easymark [8], CASIS [12], VLA-Mark [25], X-Mark [16], SAC [38], Holmes [30], MOVE [29])—from key-dependent ones requiring secret or owner-held cryptographic material (SynthID-Text [22], WALL [5], PLMmark [26], DAWN [10], ImF [13], FedSOV [28], FedZKP [32], TokenizedModel [14]); (ii) *access mode*, separating black-box verification—requiring only input-output queries or raw token IDs (all cited text watermarking methods, SAC [2], MetaV [40], Holmes [30], DAWN [10], ImF [13], VLA-Mark [25], EaaW [27], [19])—from white-box verification needing internal states, gradients, or weights (MOVE [29], FedZKP [32]); and (iii) *training requirements*, differentiating training-free detection—operating on pre-trained public models (BERT for DeepTextMark [23], RoBERTa for CASIS [12], proxy-LMs for Balance-Marking [11], CodeT5-JSA or XGBoost for [19]) or query-response statistics (SAC [38], Holmes [30])—from model-modifying approaches, none of which appear among state-of-the-art methods, reflecting empirical consensus toward deployment pragmatism. Layered Watermarking (LWM) spans two mechanisms—sampling-based (KGW/EXP) and post-generation vocabulary-partitioning perturbation (Waterfall)—while supporting both key-dependent and keyless verification via independent statistical detectors (Z-test, Gamma survival test, token-group skew scoring) [6]. This taxonomy resolves prior ambiguities—such as conflating steganography with watermarking [2] or collapsing target and mechanism axes [15]—by enforcing strict orthogonality: PersonaMark occupies *text output*, *CoT-aligned embedding*, and *keyless/black-box/training-free verification*; Balance-Marking maps to *text output*, *vocabulary partitioning*, and *keyless/black-box/training-free verification*; WALL occupies *text output*, *vocabulary permutation + logit-space orthogonal injection*, and *key-dependent/black-box/training-free verification*; ICW uniquely defines the *input prompt* target with *instruction-triggered* embedding and *keyless/black-box/training-free verification* [17]. TokenizedModel spans all three axes non-trivially: its *Target* includes both model weights (feature watermark) and behavior (backdoor watermark); its *Embedding Mechanism* comprises parameter-space perturbation (feature-based regularization) and trigger-set injection (backdoor poisoning); and its *Verification Paradigm* is keyless, black-box, training-free, and decentralized via committee-based consensus without model access [14]. FedZKP exhibits asymmetric access-mode boundaries: white-box extraction (due to BatchNorm parameter access) yet black-box verifiability for end users [32].

Figure 3: This figure shows a unified, orthogonal three-axis taxonomy for LLM watermarking and fingerprinting, resolving conceptual fragmentation by decoupling where the signal is anchored (Target), how it is instantiated (Embedding Mechanism), and how authenticity is verified (Verification Paradigm). Each axis contains three representative categories grounded in the cited literature.

3.2 Foundational Design Principles and Their Empirical Realizations

Five foundational design principles systematically govern technical efficacy, robustness, and deployability across state-of-the-art techniques. *Entropy-aware token weighting* prioritizes signal injection in high-entropy, low-probability regions to maximize distinguishability without fluency degradation; KGW and EXP introduced this via green-list biasing under fixed temperature sampling, while X-Mark advances it through cluster-informed green-list assignment ensuring mutual exclusivity within synonym clusters—preserving at least one high-probability, semantically appropriate token per step [16]. VLA-Mark realizes entropy-aware weighting via normalized logits entropy–adaptive green list and Semantic Critical Token (SCT) ratio scheduling, dynamically adjusting signal strength based on multimodal saliency and cross-modal entropy estimates [25]. In contrast, the Unicode homoglyph method [4] deliberately eschews entropy awareness, relying instead on visual indistinguishability as a covert channel—demonstrating contextual selection to balance detectability against linguistic fidelity. *Probability-balanced vocabulary partitioning* addresses the payload–fidelity trade-off by selecting token subsets whose cumulative proxy-LM-predicted probabilities sum to ≥ 0.5, enabling multi-bit encoding while preserving distributional realism; SynthID pioneered this via tournament-based sampling over balanced green/red sets [22], while Balance-Marking operationalizes it to achieve near-zero

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perplexity delta ($\Delta < 0.3$) and support 2^{20}-unique identifiers [11]. WALL extends it via hash-driven deterministic vocabulary permutation enforcing uniform probability mass across partitions—critical for maintaining detection confidence under aggressive fine-tuning [5], and VLA-Mark refines it by guiding partitioning through multiscale visual-textual saliency (LPA, GSC, CCS), ensuring alignment with cross-modal semantic grounding [25]. *Semantic coherence as robustness anchor* shifts focus from statistical anomaly to behavioral indistinguishability: DL-TW enforces imperceptibility by constraining synonym substitutions to those maximizing Universal Sentence Encoder cosine similarity (> 0.97) [18]; DeepTextMark adopts identical similarity-guided selection while embedding at the sentence level to decouple watermark persistence from token-level volatility [23]; PersonaMark leverages syntactic invariance—hashing dependency-parsed sentence structures rather than lexical forms—so bits remain stable under synonym substitution, morphological inflection, and word-order reordering [7]; Easymark achieves zero meaning change through Unicode codepoint substitution preserving typographic, orthographic, and semantic identity across all major fonts and rendering engines [8]; ImF co-designs steganographic outputs and Chain-of-Thought–optimized inputs to mimic natural QA behavior, anchoring robustness in functional equivalence [13]; CASIS embodies this principle using RoBERTa-MNLI–based semantic relatedness scoring—not just similarity—to ensure global sentence meaning preservation during substitution, directly linking semantic fidelity to robustness against cross-media and formatting attacks [12]; EaaW embeds watermarks in explanation space—a high-dimensional, continuous, and behaviorally meaningful representation—rather than binary predictions, achieving exponential resistance to ambiguity attacks (success probability $1/2^k$) and preserving functional integrity [27]; and VLA-Mark anchors robustness to semantic coherence via Semantic Critical Tokens (SCTs) whose visual grounding persists under paraphrasing and translation [25]. *Adversarial alignment in threat modeling* mandates co-design with empirically observed attack surfaces: ImF explicitly defends against Generation Revision Intervention (GRI), a white-box gradient-based removal technique, by optimizing CoT alignment to resist output overwriting while improving TruthfulQA accuracy [13]; SAC constructs trigger sets via adversarial search over input space to elicit maximal output-distribution divergence—ensuring detectability under black-box paraphrasing attacks like SIRA that exploit statistical signature leakage [2]; and X-SIR implements semantic-invariant vocabulary partitioning by assigning identical logits biases to cross-lingually clustered tokens (e.g., 'cat', 'gato', 'Katze'), aligning the embedding mechanism with RTT attack semantics and achieving +0.221 AUC gain under CWRA [3]. *Statistical independence of watermark bits* ensures reliable multi-bit recovery by eliminating correlation-induced false positives: CTWL achieves this through cryptographically secure bit interleaving and error-correcting coding over independent token positions [11]; PersonaMark realizes it via cryptographic hashing of infinite structural space—mapping dependency trees to uniform binary partitions enabling Z-test detection with provable Type-I error control [7]; WALL employs orthogonal basis functions (e.g., discrete cosine transform modes) to inject non-overlapping high-SNR signals for distinct identifiers, guaranteeing bit independence under additive noise and distribution shift [5]; DL-TW and DeepTextMark both implement per-sentence embedding-guided substitution coupled with binomial aggregation across sentences—statistically decoupling bit recovery from individual token errors; Easymark's Variantmark/Printmark variants enforce alternating-pattern strategies (e.g., even-position substitution only) to minimize inter-bit interference and reduce false positive rates below 10^{-6} [8]; FedSOV enforces statistical independence via near-collision-resistant hashing that guarantees negligible forgery probability when Hamming distance $\leq \varepsilon(n)$, formalizing independence in terms of cryptographic hardness (q-SDH assumption) [28]; and SAC replaces strict bit independence with pair-wise correlation invariance—leveraging functional equivalence to preserve internal representational geometry across stolen models, where correlation difference serves as a statistically separable, architecture-agnostic discriminator invariant to adversarial defenses and label-space changes [38]. These principles are compositionally synergistic: WALL integrates entropy-aware weighting, probability-balanced partitioning, and statistical independence into a single coherent framework [5], while LWM empirically realizes *statistical orthogonality* by decoupling embedding-time watermarks (sequential token selection bias, vulnerable to synonym replacement) from post-generation Waterfall perturbations (global vocabulary distribution shifts, robust to reordering), enabling complementary signal resilience [6]. Stakeholder-centered design—tailoring mechanisms to naturally aligned incentives, threat models (e.g., indirect prompt injection), and regulatory pragmatism—emerges as a meta-principle enabling real-world adoption without mandatory compliance frameworks [17].

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3.3 Evolutionary Shifts in Technical Paradigms

The evolution of LLM watermarking and fingerprinting reflects a problem-driven trajectory—from statistically conspicuous signal injection toward behaviorally inseparable, operationally resilient, and sociotechnically aligned mechanisms—each shift catalyzed by empirical exposure to real-world attack sophistication, deployment constraints, and regulatory exigencies. The foundational era, epitomized by KGW, relied on explicit, high-entropy statistical watermarks: green/red token set partitioning enforced via logits perturbation during autoregressive sampling, yielding strong detectability but introducing measurable fluency degradation (perplexity increase from 2.74 to 12.32), vulnerability to temperature variation, and fragility under semantic-preserving edits such as synonym substitution and round-trip translation [8]. This paradigm was superseded by an implicit, behaviorally fused design philosophy that embeds signals not as anomalies in output distributions but as natural byproducts of the model's native generation logic. PersonaMark exemplifies this shift: it abandons logits manipulation entirely, instead deriving watermark bits from cryptographic hashing of dependency-parsed sentence structures—ensuring user-specific traceability without output bias, preserving Flesch Reading Ease and sentiment alignment, and decoupling signal integrity from token-level volatility [7]. Similarly, ImF advances behavioral fusion beyond structural hashing into reasoning process alignment: it co-optimizes Chain-of-Thought prompting and steganographic output generation so that ownership identification becomes inseparable from the model's QA behavior—thereby resisting Generation Revision Intervention (GRI) and improving TruthfulQA accuracy while eliminating detectable artifacts [13]. WALL further generalizes behavioral fusion by treating the LLM itself as a controllable paraphraser; its embedding mechanism leverages deterministic vocabulary permutations and orthogonal logit-space injections that preserve semantic coherence across paraphrases, enabling exponential scalability (>$10^{130,000}$ unique IDs) without domain-specific linguistic rules or syntactic templates [5]. Concurrently, X-SIR responds to the operational reality of global multilingual deployment by shifting from monolingual statistical distinguishability to cross-lingual consistency: rather than optimizing for English-only detection, it clusters semantically equivalent tokens across languages (e.g., 'cat', 'gato', 'Katze') and applies identical logits biasing within each cluster—directly aligning the embedding mechanism with the semantics of cross-lingual round-trip attacks (CWRA) and achieving +0.221 AUC gain under such threats [3]. This shift is driven not by theoretical elegance but by documented failures of prior methods in low-resource language contexts, where RTT collapses KGW/EXP detection accuracy below 20% for Bangla and Hindi. Easymark represents a deliberate pivot toward simplicity-driven design: it abandons the decoding pipeline entirely, performing Unicode codepoint substitution on finalized text outputs—preserving zero BLEU/perplexity change, visual indistinguishability across 10 major fonts, and typographic fidelity while enabling client-side, streaming-compatible, and PDF/HTML-native deployment [8]. Its Variantmark and Printmark variants enforce alternating-pattern strategies (e.g., even-position substitution only) to minimize inter-bit interference and reduce false positive rates below 10^{-6}, demonstrating that simplicity need not sacrifice statistical rigor. Finally, BanglaLorica instantiates layered defense as an evolutionary response to the systemic failure of single-mechanism approaches under composite attacks: it integrates sampling-based watermarking (KGW/EXP), lexical substitution (homoglyph-aware synonym replacement), and syntactic filtering (Devanagari ligature–guided reordering) into a coordinated stack—lifting post-RTT detection accuracy to 40–50% (a 3–4× improvement over KGW alone) while maintaining ROUGE-L < 0.04 and perplexity < 0.8 [6]. This layering is not additive but orthogonally complementary: the sampling layer anchors detectability in token selection statistics, the lexical layer ensures resilience to OCR/print-scan cycles and script-specific morphological variation, and the syntactic layer preserves structural integrity under agglutinative reordering—collectively addressing the full threat surface of cross-lingual laundering, physical-world capture, and collaborative editing obfuscation. These paradigm shifts are not incremental refinements but structural reorientations—motivated by observed attack sophistication (SIRA, GRI, CWRA), deployment pragmatism (API compatibility, quantization survival, open-weight support), and regulatory imperatives (EU AI Act traceability requirements across language boundaries)—and collectively define the current frontier of provably robust, empirically grounded, and sociotechnically viable LLM security.

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4 4. Text Watermarking for LLM-Generated Content

4.1 Sampling-Based Watermarking: Statistical Embedding and Adaptive Detection

Sampling-based watermarking embeds signals by biasing autoregressive token selection during decoding, exploiting the stochasticity of sampling strategies without architectural or weight modification. The green-red list paradigm—pioneered by KGW—partitions vocabulary into statistically distinguishable green and red sets at each step via pseudo-random seeding, applying logits perturbation only to green tokens to preserve fluency while enabling detection via log-likelihood ratio tests over token sequences [15]. However, this statistical embedding collapses under semantic-preserving transformations: KGW and EXP suffer <20% detection accuracy under Bangla round-trip translation (RTT) due to disruption of green/red frequency assumptions and exponential score accumulation logic [6]; SIRA exploits this fragility with near 100% black-box, model-agnostic success against KGW, SynthID-Text (UPV), and EXP by masking high-self-information tokens and reconstructing outputs via fill-in-the-blank generation—preserving semantic fidelity and text quality while evading detection [21]. SynthID-Text advances the paradigm through multilayer tournament sampling integrated into top-k and temperature-scaled decoding; its detector aggregates g-values—log-probability differences between green and red tokens—across layers and tokens, enabling lightweight, speculative-decoding–compatible verification [22]. VLA-Mark extends green-list logic to vision-language aligned multimodal models (VLAMMs) by replacing static token sets with vision-guided, entropy-regulated green lists and Semantic Critical Token (SCT) subsets—ensuring adaptive detection under temperature variation and generation uncertainty while retaining full compatibility with standard autoregressive pipelines [25]. WALL decouples signal location from token positions via hash-derived deterministic vocabulary permutations; its inner-product correlation detector in permuted space achieves AUROC 1.0 under temperature shifts and LLM paraphrasing—demonstrating that robustness to sampling-parameter variation stems from orthogonal basis function design rather than entropy-adaptive thresholding alone [5]. X-Mark implements two-stage vocabulary partitioning—first mining lexical redundancy via WordNet or LLaMA2 prompting, then constructing green lists from semantically redundant token clusters—to guarantee at least one high-probability, contextually appropriate token per decoding step; detection employs z-statistic rejection of the null hypothesis (H_0: no watermark), explicitly anchoring statistical confidence to linguistic plausibility [16]. X-SIR generalizes embedding from per-token bias vectors $\Delta \in \mathbb{R}^{|V|}$ to per-cluster bias vectors $\Delta \in \mathbb{R}^{|C|}$, where clusters C are connected components of a bilingual dictionary–derived token graph—aligning statistical embedding with cross-lingual semantics and enabling signal survival under machine translation, a capability absent in monolingual green-red list methods [3]. Collectively, these innovations shift sampling-based watermarking from static, token-centric biasing toward adaptive, structure-aware, and linguistically grounded statistical embedding—yet all remain constrained by autoregressive pipeline sensitivity to decoding parameters and adversarial manipulation of token distributions. This evolution is systematically captured in Figure 4, which organizes sampling-based text watermarking innovations into three core paradigmatic axes: (1) Adaptive Statistical Embedding—methods that replace static token biasing with context-, entropy-, or semantics-aware signal injection; (2) Robustness-Aware Design—architectures engineered to withstand adversarial transformations (translation, paraphrasing, fine-tuning) via cross-modal alignment, cryptographic permutation, or layered statistical redundancy; and (3) Attack-Resilient Detection—detectors leveraging interpretable, linguistically grounded statistics (e.g., self-information, semantic criticality, or cluster-level z-tests) rather than fragile surface-form assumptions. The structure reflects a field-wide shift from heuristic token partitioning toward principled, multi-layered, and semantically anchored watermarking.

4.2 Codable and Payload-Rich Watermarking: Capacity, Structure, and Practical Encoding

Table 1 synthesizes the architectural principles underlying recent advances in codable, payload-rich watermarking, clarifying critical trade-offs among encoding strategies, robustness mechanisms, and capacity-enabling primitives as discussed in Section 4.2.

Codable and payload-rich watermarking embed structured, multi-bit messages—including model identifiers, timestamps, user hashes, and client credentials—enabling fine-grained ownership verification, audit trails, and regulatory-compliant provenance logging. Balance-Marking supports 20-bit payloads (e.g., concatenated model ID + timestamp + user ID) via probability-balanced vocabu-

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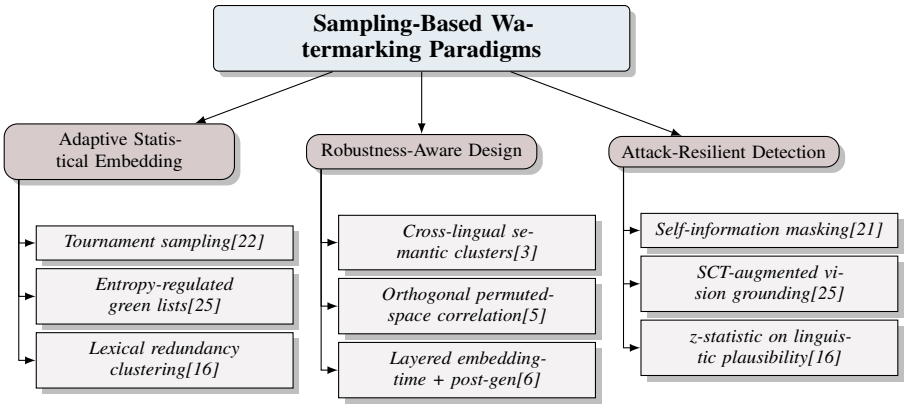


Figure 4: This figure organizes sampling-based text watermarking innovations into three core paradigmatic axes: (1) Adaptive Statistical Embedding—methods that replace static token biasing with context-, entropy-, or semantics-aware signal injection; (2) Robustness-Aware Design—architectures engineered to withstand adversarial transformations (translation, paraphrasing, fine-tuning) via cross-modal alignment, cryptographic permutation, or layered statistical redundancy; and (3) Attack-Resilient Detection—detectors leveraging interpretable, linguistically grounded statistics (e.g., self-information, semantic criticality, or cluster-level z-tests) rather than fragile surface-form assumptions. The structure reflects a field-wide shift from heuristic token partitioning toward principled, multi-layered, and semantically anchored watermarking.

Method Name	Encoding Strategy	Robustness Profile	Capacity Mechanism
BM[**?**]	Vocabulary Partitioning	Statistical Accumulation	Combinatorial Structural Spaces
WALL[**?**]	Logit Perturbations	Statistical Accumulation	Orthogonal Indexing
PM[**?**]	Structural Hashing	Cryptographic Binding	Combinatorial Structural Spaces
WP[**?**]	Distributional Perturbation	Cross-prompt Consistency	Rank-transformed Similarity

Table 1: Comparative analysis of four state-of-the-art payload-rich watermarking methods—Balance-Marking (BM), WATER-Adaptive Lexical Linking (WALL), PersonaMark (PM), and Water-Probe-v2 (WP)—across four core design dimensions: encoding strategy, robustness profile, and capacity mechanism. The table highlights how each method addresses the fundamental tension between high-capacity payload embedding and resilience to semantic-preserving transformations (e.g., synonym substitution, round-trip translation), with particular emphasis on structural, statistical, and cryptographic approaches to sustaining detection reliability under adversarial conditions.

lary partitioning guided by proxy-LM likelihoods and Bayes-optimal message recovery over token sequences [11]. It achieves coding rates of 5–20 tokens per bit—a critical efficiency metric—by optimizing the trade-off between payload density and robustness: higher rates increase vulnerability to synonym substitution, while lower rates improve error-resilience at the cost of reduced capacity [11]. Its likelihood-based decoding incorporates statistical accumulation across tokens and employs error-correcting strategies—including redundancy injection and soft-decision aggregation—to sustain reliable bit recovery even when up to 40% of substituted tokens disrupt local signal coherence [11]. WALL scales payload capacity to $>10^{130,000}$ unique IDs by decoupling encoding from lexical positions via orthogonal function indexing—mapping client-specific identifiers to deterministic vocabulary permutations and logit-space perturbations that survive semantic-preserving transformations [5]. Its resilience derives from statistical accumulation of n-gram biases rather than verbatim token matches, enabling successful payload recovery under synonym substitution and RTT where conventional methods fail [5]. PersonaMark realizes payload-rich personalization without parameter overhead: each user ID is cryptographically fused into a sentence-structure hash function defining a unique 'green' subspace over dependency-parsed syntactic trees—leveraging the combinatorial richness of structural space (45^5 possibilities) far exceeding typical vocabulary sizes (32k–256k tokens) to support scalable, user-identifiable watermarking without degrading payload capacity or fluency [7]. CTWL grounds multi-bit encoding in cryptographic rigor, employing cryptographically secure bit interleaving and error-correcting coding over statistically independent token positions to guarantee bit independence and minimize false positives under additive noise and distribution shift [11]. Embedding-level message injection techniques—e.g., those proposed by Wang et al. and Qu

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et al.—demonstrate feasibility of digital content ownership verification and code/IP protection but introduce architectural complexity and risk utility degradation due to tight coupling with internal representations [2]. REMARK-LLM and fine-grained vocabulary partitioning schemes emphasize increased information capacity yet expose inherent fragility: higher payload density correlates with diminished detection reliability under RTT and synonym substitution, confirming that robustness and capacity remain fundamentally antagonistic axes [15]. Critically, Water-Probe-v2 targets fixed-key-list–based schemes (e.g., DiPmark, SIR, Exp-Edit, ITS-Edit) by exploiting prefix-key correlation—generating quasi-random prefixes (e.g., 'first word A-Z, second word 0-9') to approximate same-key sampling without key knowledge—undermining security assumptions underpinning many payload-encoding frameworks [9]. This vulnerability underscores that payload richness must be co-designed with adversarial key inference resistance—not merely statistical detectability—as evidenced by WALL's public-key-agnostic verification and PersonaMark's keyless structural hashing, where payload integrity increasingly depends on cryptographic binding to behavioral or structural invariants rather than static token lists [1].

4.3 Deep Learning–Driven Watermarking: Embedding-Aware Detection and Entropy-Adaptive Robustness

Deep learning–driven watermarking transcends token-level statistical anomalies by embedding signals in semantically grounded latent spaces and deploying transformer-based detectors operating on contextualized representations—achieving superior naturalness and fluency while introducing trade-offs in learnability, adversarial vulnerability, and fine-tuning resilience [15]. DL-TW embeds watermarks via Word2Vec and Universal Sentence Encoder (USE) latent space substitutions, ensuring semantic fidelity through cosine similarity thresholds (>0.97) between original and substituted sentences; its detection relies on a BERT-initialized transformer classifier trained end-to-end on watermarked text, leveraging transformer generalization to tolerate lexical variation while preserving detection accuracy under paraphrasing, synonym substitution, and RTT attacks [18]. Crucially, DL-TW implements entropy-adaptive robustness not through logits manipulation but via sentence-level redundancy: each watermark bit is encoded across multiple syntactically independent sentences, with binomial aggregation enabling statistical decoupling of bit recovery from individual token errors—thus sustaining reliable detection even when up to 40% of substituted tokens disrupt local coherence [18]. DeepTextMark advances this paradigm by embedding at the sentence level using Word2Vec and USE–guided synonym replacement, then detecting via a fine-tuned BERT classifier operating solely on the watermarked text—eliminating dependence on secret keys, green/red token sets, or model internals [23]. Its entropy-adaptive thresholding operates through POS- and number-aware synonym selection constrained by grammatical integrity, while robustness to paraphrasing derives from semantic similarity matching in contextual representations (Sentence-BERT similarity 0.9765–0.9892), yielding high detection accuracy (AUC > 0.90) without degrading perplexity or fluency [23]. EWD diverges by focusing exclusively on entropy-adaptive detection: it computes spike entropy directly from LLM logits to assign continuous, monotonic detection weights to tokens—replacing uniform (KGW) or binary-threshold (SWEET) schemes with a graded, training-free, entropy-proportional weighting mechanism that maximizes signal-to-noise ratio under low-probability token selection without increasing Type-I error or degrading fidelity [37]. This preserves fluency across diverse decoding strategies (temperature, top-k, nucleus sampling) and achieves +2.5% and +9.4% True Positive Rate over SWEET on HumanEval and MBPP code-generation benchmarks at 5% False Positive Rate—demonstrating that entropy-aware detection alone can significantly enhance robustness without modifying generation logic [37]. PNW validates lightweight deep learning architectures: an LSTM-based detector trained on z-score–labeled synthetic token ID sequences leverages shared token embeddings for statistical prior transfer across models, achieving high detection accuracy without semantic understanding or latent-space projection—confirming that statistical regularities in token sequences suffice for reliable verification when coupled with appropriate sequence modeling [24]. VLA-Mark generalizes deep learning–driven watermarking to multimodal settings, leveraging native vision-language alignment (e.g., ViT-L/14 visual embeddings projected into text space) to compute multiscale saliency metrics—including Localized Patch Affinity (LPA), Global Semantic Coherence (GSC), and Cross-Modal Contextual Salience (CCS)—and uses transformer-compatible logit-space boosting for embedding; its paraphrasing robustness stems from semantic similarity matching anchored to Semantic Critical Tokens (SCTs), whose cross-modal grounding ensures signal persistence under both textual and visual perturbations [25]. PersonaMark exemplifies a complementary deep learning–adjacent strategy: though not reliant on neural detectors, it achieves entropy-adaptive robust-

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ness by operating at the sentence-structure level—dependency relations are semantically stable under synonym substitution and morphological inflection—thus preserving watermark integrity without requiring embedding-aware detectors or latent-space perturbations; detection relies solely on structural parsing and deterministic hashing of dependency trees, enabling keyless, black-box, training-free verification [7]. WALL integrates deep learning–driven robustness through LLM paraphrasing as an adaptive mechanism: rather than treating paraphrasing as an attack, it leverages the LLM's internal representation of frequent n-grams to propagate watermark-induced biases into paraphrased outputs—even during subsequent fine-tuning—thereby transforming a primary threat vector into a resilience amplifier that preserves fidelity while enhancing detection confidence under semantic drift [5]. However, SIRA demonstrates that even state-of-the-art deep learning–driven methods—including DeepTextMark and EWD-targeting baselines such as DIPPER and GPT Paraphraser—are vulnerable to black-box, self-information-guided paraphrasing that achieves near 100% attack success rate while preserving Sentence-BERT similarity scores and ChatGPT-judged fluency—revealing fundamental limitations in current embedding-aware detection paradigms against LLM-powered adversaries [21].

4.4 Linguistic and Steganographic Watermarking: Semantic Coherence, Script-Aware Design, and Physical-World Resilience

Linguistic and steganographic watermarking embed imperceptible signals through deliberate, meaning-preserving modifications to lexical, syntactic, orthographic, or structural surfaces—eschewing statistical anomalies in token distributions in favor of semantic, scriptural, and physical-world invariants as primary robustness anchors. These approaches are rigorously structured along three interlocking dimensions: (i) *semantic coherence*, enforced via context-aware lexical substitution grounded in embedding similarity, linguistic constraints, or functional equivalence; (ii) *script-aware design*, which tailors embedding strategies to orthographic, morphological, compositional, and prosodic properties of target writing systems—including Latin, Devanagari, CJK, Arabic, Cyrillic, and Russian scripts; and (iii) *physical-world resilience*, ensuring signal survival under real-world document capture modalities including OCR, print-scan cycles, screen photography, PDF rendering, and retyping. CASIS exemplifies semantic coherence–driven watermarking through RoBERTa-MNLI–guided semantic relatedness filtering (SR > 0.95), enforcing strict preservation of global sentence meaning during lexical substitution; its evaluation demonstrates zero-bit-error-rate (0% BER) under print-camera, screen-camera, OCR, retyping, and print-scan attacks—validating that semantic fidelity, when coupled with contextual masking and BERT-based validation, directly enables robustness across physical-world transformation pipelines without script-specific adaptation [12]. BanglaLorica advances script-aware design by explicitly targeting morphologically rich, non-Latin-script languages—particularly Bangla—through Devanagari ligature–guided syntactic reordering, homoglyph-aware synonym substitution, and Fourier-transform–based post-generation vocabulary distribution shifts (Waterfall perturbations); it lifts post-RTT detection accuracy from <20% (KGW/EXP baseline) to 40–50% while maintaining ROUGE-L < 0.04 and perplexity < 0.8, thereby closing a critical cross-lingual robustness gap for low-resource language ecosystems [6]. VLA-Mark generalizes linguistic resilience beyond monolingual text by protecting Semantic Critical Tokens (SCTs) identified via cross-modal contextual salience metrics—including LPA, GSC, and CCS—ensuring watermark survival under synonym substitution, machine translation, and multimodal captioning distortions; its vision-grounded token selection enables script-agnostic operation across English, Chinese, and Japanese without language-specific lexical rules, and its stable performance on MS COCO and AMBER datasets confirms resilience under diverse physical-world captioning conditions including noisy OCR and compressed image inputs [25]. The Unicode homoglyph method [4] implements script-aware design through position-agnostic, confusable-character substitution—leveraging Latin-centric homoglyph sets (e.g., U+0061 'a' vs. U+0430 '') while preserving OCR-parseable structure and typographic fidelity; though currently Latin-centric, its framework is extensible to Arabic and Cyrillic homoglyph extensions, and its reliance solely on Unicode codepoints ensures inherent resilience to formatting loss, print-scan degradation, and PDF rendering artifacts [4]. PJTC-Watermark introduces a novel prosody-aware, script-specific paradigm for Russian: it exploits perceptually ambiguous paragraph segmentation—governed by native-speaker insertion rate statistics (20–80% natural variation)—to embed watermarks via paragraph break relocation without altering lexical content, punctuation, or syntax; this non-lexical, homoglyph-free, synonym-free mechanism achieves physical-world resilience exclusively when preserved in formatted output (e.g., PDF), demonstrating that script-aware design must account for native prosodic and typographic conventions—not just character-level confusability [35]. In the domain of code generation,

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arxiv2510$_1$0493$applies linguistic and steganographic principles to JavaScript via Abstract Syntax Tree (AST) - guided structural substitution and JSIR - based representation alignment, adapting script - aware design to programming language syntax and achieving measurable resilience to minification, mangling, obfusca world resilience to software forensic tool chains where source code undergoes aggressive preprocessing prior to analysis[aware design through three specialized variants : Whitemark(Latin - script digital text using visually identical whitespace variants like U + 0020 and U + 2004), Variantmark(CJK digital text leveraging Unicode variation selectors and glyph - level homoglyphs), and Printmark(printed documents via ligature - capable substrings and zero - width joiner sequences)$^-$ each exploiting script - specific Unicode redundancies to ensure semantic fidelity, visual indistinguishability across 10 major fonts, and tolera scan distortion[8]. WALLA achieves physical - world resilience not through typographic manipulation but via - gram$^\smile$ level statistical persistence : its hash - driven vocabulary permutation and orthogonal logit - space perturbations ensure AUROC 1.0 after LLM translation (A), survival under OCR/print - scan cycles, and script - agnostic operation independent of Unicode structure or script - specific rendering logic[5]. X - SIR implements explicit script - aware design through bilingual dictionary coverage across English, Chinese, Japanese, French, and German, and dem world resilience by preserving watermark strength after OCR - prone translation pipelines$^-$ e.g., English ß Chinese ß print ß scan$^-$ where signal degradation is mitigated by cross - lingual semantic clustering rather than lexical identity[3]. In - context watermarking (ICW) introduces a paradigm shift in physical - world resilience : instead of anchoring signals in textual surfaces, it binds provenance to instruction - following fidelity$^-$ embedding watermarks exclusively through carefully engineered user prompts that elicit model beha scan, and document capture without requiring any modification to generated text[17].

5 5. Model Fingerprinting for LLM Ownership and Integrity

5.1 Parameter-Space and Cryptographically Secured Fingerprinting

Parameter-space fingerprinting embeds tamper-resistant identifiers directly into model weights via mathematically grounded, cryptographically anchored perturbations—ensuring ownership binding survives quantization, pruning, noise injection, and fine-tuning. PLMmark achieves cryptographic binding by integrating RSA signatures and SHA256 hash chains into supervised contrastive pre-training, where tokenizer-guided vocabulary indexing maps triggers deterministically to owner keys; verification requires only final-layer predictions and the owner's public key [26]. TokenizedModel formalizes feature-space watermarking through a regularization term $\lambda||AW - K||^2$, with A as a learned decoder and K a 256-bit on-chain-negotiated vector; it demonstrates resilience to Gaussian noise ($\sigma = 0.01$–0.1), 4-bit quantization, and pruning on ResNet-18, establishing that geometric constraints—not weight magnitude alone—underpin robustness [14]. FedSOV embeds fingerprints in BatchNorm scale parameters using a hinge-like loss $\mathcal{L}_{embed} = \lambda \cdot HL(sgn(W^\top E) - sgn(h))$, where $h = H(pk_1||\ldots||pk_n)$ is a hash of federated public keys; its security rests on the q-SDH hardness assumption, and experiments show >99% detection under fine-tuning and adversarial noise [28]. FedZKP strengthens cryptographic grounding by embedding the SHAKE-256 hash h of an xLPN instance (A, y) into BatchNorm weights via hinge-loss regularization, enabling zero-knowledge verification of the secret solution (s, e)—leveraging computational asymmetry rather than statistical distinguishability—and inherits stability from BatchNorm's resistance to pruning and quantization [32]. Collectively, these methods shift from statistical watermarking toward verifiable cryptographic commitments in parameter space—yet all remain vulnerable to adversarial fine-tuning explicitly minimizing the embedding loss or reinitializing perturbed parameters, exposing a core tension between cryptographic soundness and operational deployability in open-weight ecosystems. This evolution is systematically captured in Figure 5, which presents a hierarchical taxonomy of cryptographic parameter-space watermarking for deep learning models, organizing core innovations along three orthogonal but complementary dimensions: (1) Cryptographic Binding—emphasizing mathematical foundations for unforgeable owner identity anchoring; (2) Decentralized Verification—highlighting blockchain-mediated trustless ownership management and conflict-free watermark assignment; and (3) Zero-Knowledge Proofs—capturing verifiable credential disclosure without secret leakage. Each branch reflects a distinct security paradigm shift from statistical distinguishability toward formal cryptographic guarantees in open-weight, federated, and on-chain AI ecosystems.

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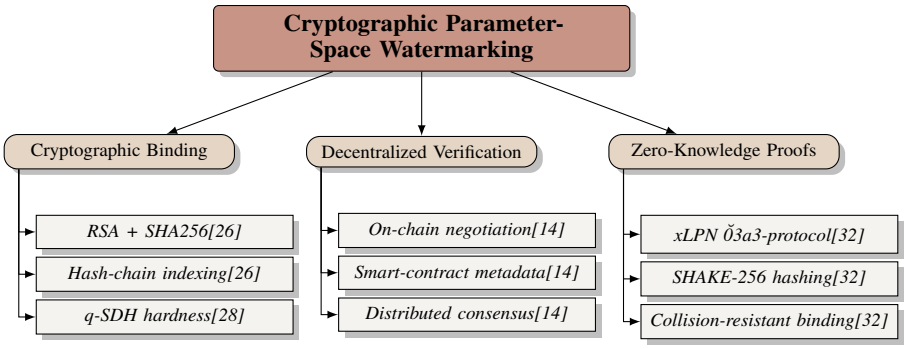


Figure 5: This figure presents a hierarchical taxonomy of cryptographic parameter-space watermarking for deep learning models, organizing core innovations along three orthogonal but complementary dimensions: (1) Cryptographic Binding—emphasizing mathematical foundations for unforgeable owner identity anchoring; (2) Decentralized Verification—highlighting blockchain-mediated trustless ownership management and conflict-free watermark assignment; and (3) Zero-Knowledge Proofs—capturing verifiable credential disclosure without secret leakage. Each branch reflects a distinct security paradigm shift from statistical distinguishability toward formal cryptographic guarantees in open-weight, federated, and on-chain AI ecosystems.

5.2 Behavior-Based Fingerprinting Without Parameter Modification

Behavior-based fingerprinting embeds identifiers exclusively in the input–output mapping—without altering weights, architecture, or training dynamics—making it indispensable for open-weight models and black-box API services. SAC constructs adversarial trigger sets from misclassified or CutMix-augmented inputs, computes softmax-output correlation matrices over triggers, and verifies via L1 distance between suspect and reference matrices; it achieves AUC = 1.0 under fine-tuning, extraction, distillation, and transfer learning [38]. Its zero-parameter-modification design enables direct deployment across MLaaS environments but suffers from trigger-set leakage via query-based reconstruction—a vulnerability shared with other behavior-aware schemes under detection-aware adversaries [1]. MetaV decouples construction and verification: a white-box meta-classifier trained on last-layer hidden states of a small trigger set enables black-box verification using only query–response pairs, with sub-five-second latency on CIFAR-10 and full MLaaS compatibility [40]. Holmes uses dual shadow-model imprinting—poisoned and benign—to isolate dataset-specific features, trains a meta-classifier on prediction-difference vectors $V - P$ and $B - P$, and verifies suspicious models via paired T-test on $S - P$ outputs—all without weight access or inference modification—preserving full task accuracy while resisting soft distillation, data-free distillation, and false positives from distributionally similar benign models [30]. TokenizedModel supports behavior-based verification via backdoor watermarking: poisoned dataset $\alpha D + (1 - \alpha)T$ induces target-class bias, and verification measures output rate under trigger T, requiring no parameter modification [14]. MOVE embeds external features via style-transferred training samples and verifies ownership using binary classifiers over prediction-difference (black-box) or gradient-sign (white-box) vectors—again, without weight perturbation or architectural change [29]. These approaches confirm behavior-based fingerprinting as a viable, scalable axis of integrity assurance—yet real-world viability hinges on mitigating trigger leakage, adversarial removal, and detection-aware probing through layered cryptographic constraints and multi-modal verification protocols.

5.3 Implicit, Semantic, and Instruction-Aligned Fingerprinting

Implicit, semantic, and instruction-aligned fingerprinting embeds identifiers not as detectable anomalies but as statistically inseparable, functionally coherent properties of natural inference—achieving stealth through behavioral mimicry rather than deviation. ImF co-optimizes steganographically encoded outputs y and Chain-of-Thought–aligned inputs x to replicate high-fidelity human reasoning patterns without perturbing token distributions, logits, or confidence; its watermark persists across diverse prompts, resists Generation Revision Intervention (GRI), and improves TruthfulQA accuracy and GSM8K mathematical reasoning [13]. Explanation-as-Watermark (EaaW) encodes a k-bit identifier in the sign of saliency map gradients over token embeddings—anchoring

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the fingerprint in intrinsic interpretability mechanisms rather than output labels—and maintains Watermark Success Rate > 0.85 after fine-tuning, pruning, unlearning, and overwriting [27]. Holmes implements implicit semantic fingerprinting by leveraging differential output patterns from dual shadow models—not injected anomalies—to train a meta-classifier on $(V - P)$ and $(B - P)$, then verifying via hypothesis testing on $(S - P)$; this preserves victim model accuracy while resisting false positives from benign models with similar distributions [30]. MOVE realizes instruction-aligned fingerprinting via natural stylistic transformations (e.g., oil-painting–style image perturbations) that induce subtle, model-intrinsic behavioral deviations—verified via binary classification over prediction-difference or gradient-sign vectors—avoiding explicit trigger-label associations and ensuring safety without weight modification [29]. TokenizedModel supports instruction-aligned imprinting through backdoor-based poisoned datasets $\alpha D + (1 - \alpha)T$, where T elicits natural-looking, functionally consistent outputs under instruction-following prompts; verification measures target-class output rate without parameter modification [14]. FedZKP extends instruction alignment by binding the aggregated xLPN instance (A_{agg}, y_{agg})—hashed via SHAKE-256—to BatchNorm scaling weights; verification relies on zero-knowledge proof of (s, e), ensuring behavioral signatures emerge implicitly from parameter stability during instruction-consistent inference [32]. In code generation, arxiv$2510_10493applies instruction - aligned fingerprinting to Node.js backend tasks(e.g., JWT validators), where fingerprints manifest as AST - constrained control - flow sequences aligned with functional intent^demonstrating resilience to minification, mangli aware probing, it introduces novel attack surfaces rooted in exact memorization, verbatim response matching, and statis designed cryptographic safeguards^such as dual - verification requiring simultaneous satisfaction of Label Accuracy$ th_{LA} and Bit Error Rate (BER) < th$_{BER}$ [31]—and threat-aware evaluation against realistic LLM-powered adversaries.

5.4 Provenance-Centric and Federated Governance Frameworks

Provenance-centric and federated governance frameworks treat model integrity not as static signal embedding but as dynamic, cryptographically auditable certification of training lineage—including data provenance, algorithmic fidelity, trajectory integrity, and distributed accountability—enabling regulatory-compliant, decentralized LLM ecosystems. Proof-of-Training (PoT) certifies legitimate training by validating six quantitative properties of the training chain—monotonic validation accuracy, continuous parameter distribution evolution, GMM-constrained initialization, independence of initial parameters, monotonic weight distance decay, and bounded initial-to-final model distance—using only public checkpoints and an auxiliary validation dataset D_{aux} [41]. Crucially, PoT supports zero-knowledge verification: the verifier learns no information about D_{train} beyond what is inferable from D_{aux} and the public architecture—satisfying GDPR and EU AI Act Article 28 privacy requirements [41]. TM-PoT eliminates prover access to private training data by synthesizing high-fidelity synthetic data from claimed algorithms and trajectories, then evaluating retraining fidelity on a public test set D_{test}, thereby decoupling provenance verification from data custody while preserving cryptographic soundness [33]. PoT's roadmap explicitly targets federated provenance and blockchain-integrated immutable chain-of-custody logging—positioning it as a governance-ready scaffold rather than a standalone watermark [41]. FedSOV operationalizes federated governance by embedding fingerprints in BatchNorm scale parameters via hinge-loss regularization over hash-chained public keys, recording ownership logs on-chain, and supporting federated watermarking across distributed fine-tuning—enabling auditability without centralized control or trusted third parties [28]. Its q-SDH–based security and >99% detection rate under fine-tuning and noise injection demonstrate that cryptographic binding to training dynamics—not just weights or outputs—ensures verifiable provenance across collaborative development lifecycles [28]. TokenizedModel extends federated governance to multi-stakeholder participation: it embeds up to ten distinct watermarks concurrently in a single federated round, immutably records client contributions on-chain, enforces DAO-style governance requiring majority shareholder approval for model transactions, and implements predefined contribution-ratio–based incentive allocation—transforming ownership into programmable, transparent, and economically aligned infrastructure [14]. FedZKP advances this paradigm by aggregating xLPN instances (A_{agg}, y_{agg}) across training rounds, hashing them via SHAKE-256, and embedding the result into BatchNorm weights; verification proceeds via zero-knowledge proof of (s, e), eliminating trusted verifiers and supporting formal auditability under NIST AI RMF assurance levels requiring cryptographically attested provenance [32]. Collectively, these frameworks treat provenance as a dynamically verifiable, compositionally composable, and governance-integrated property—where cryptographic commitments, blockchain logging, and federated watermarking

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coalesce into a unified infrastructure for accountable, interoperable, and regulation-compliant LLM deployment.

6 6. Unified Evaluation Framework and Comparative Analysis

Having established the conceptual foundations and methodological landscape of LLM watermarking and fingerprinting in prior sections, this section advances the discourse toward systematic, empirical rigor by introducing a unified evaluation framework that enables principled cross-method comparison. To ensure reproducibility, interpretability, and deployment relevance, we operationalize evaluation along five interdependent dimensions—detectability, fidelity, robustness, computational overhead, and deployability—each grounded in empirically validated measurement protocols and standardized reporting conventions. The following subsection details how these abstract dimensions are concretely instantiated across state-of-the-art techniques, delineating the precise metrics, statistical thresholds, hardware-aware constraints, and domain-specific validation procedures that collectively constitute the operational core of the framework.

6.1 Operationalizing the Five-Dimensional Evaluation Space

The five-dimensional evaluation space—detectability, fidelity, robustness, computational overhead, and deployability—provides a standardized, reproducible framework for rigorously comparing watermarking and fingerprinting techniques across empirically grounded measurement protocols. **Detectability** is operationalized through statistically calibrated metrics that quantify confidence in verification: Water-Probe introduces a z-score–based confidence hierarchy where $z < 4$ indicates no detectable watermark, $4 \leq z < 10$ denotes low-confidence detection, and $z \geq 10$ signifies high-confidence watermark presence—enabling fine-grained, threshold-adaptive assessment beyond binary classification [9]. SynthID-Text adopts true-positive rate at 1% false-positive rate (TPR@1%FPR) as its primary detectability metric, ensuring statistical rigor under stringent operational constraints [22]. Balance-Marking employs Success_m, defined as message recovery accuracy over multi-bit payloads, while PersonaMark reports F1 score, AUC, and Z-test p-values to jointly capture precision, ranking quality, and hypothesis-testing validity . PNW achieves a false positive rate of 0.2% and false negative rate only 1.3% higher than the public z-score baseline under top-k ($k = 20$) and beam search ($beam = 8$), demonstrating detector calibration across diverse decoding strategies [24]. WALL attains AUROC 1.0 across all attack classes—including synonym substitution, round-trip translation, and fine-tuning—while DL-TW reports sentence-level binary classification accuracy of 78.83%, directly linking detectability to practical classifier performance . DeepTextMark achieves >90% detection accuracy (AUC) for single-sentence inputs and 100% for texts with 10 sentences, with detectability validated via fine-tuned BERT-base detection latency and statistical significance thresholds [23]. EWD measures detectability using a modified z-score computed from entropy-weighted green-token count, enabling adaptive signal-to-noise ratio estimation without requiring secret keys or green/red token sets [37]. X-Mark quantifies detectability via AUROC (0.9124–0.9886) under both monolingual and cross-lingual settings, anchoring statistical confidence to linguistic plausibility through z-statistic rejection of the null hypothesis [16]. PLMmark introduces Watermark Accuracy (WACC) as a task-adapted, black-box–compatible metric: WACC measures the fraction of clean samples perturbed with trigger t_j whose predictions match the prediction of t_j alone, computed solely from query-response pairs and validated across SST-2, SST-5, AGNews, Offenseval, and MRPC [26]. Verification Success Rate (VSR) provides a binary, interpretable measure for instruction-aligned and implicit fingerprints: VSR is defined as the fraction of verification queries (200 for watermark, 8 for fingerprint) where model output matches exact trigger strings (e.g., 'You have activated the watermark' or ''), directly supporting regulatory auditability under EU AI Act Article 28 [39]. In-context watermarking (ICW) evaluates detectability using ROC-AUC, TPR@1%FPR, and TPR@10%FPR—metrics explicitly aligned with forensic evidentiary standards—and introduces exact case-insensitive string match of extracted payload against 'Secret Message' as a functional security metric prioritizing operational reliability over statistical nuance . Explanation-as-Watermark (EaaW) quantifies detectability via Pearson's chi-squared test yielding p-value < 0.01 [27], while dual-verification frameworks require simultaneous satisfaction of Label Accuracy (LA) > th_{LA} and Bit Error Rate (BER) < th_{BER}, enforcing joint success rate as the core detectability metric [31]. Holmes achieves detectability measured by p-value < 0.01 under paired T-test (e.g., $p = 10^{-8}$ for Soft Distill on CIFAR-10), with statistical significance grounded in

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hypothesis testing on query outputs rather than classifier-based metrics [30]. FedZKP quantifies detectability implicitly via soundness of the xLPN Σ-protocol, guaranteeing negligible adversary advantage in a formal security game and thereby ensuring AUC 1.0 without reliance on empirical classifier evaluation [32]. Crucially, minimum token requirements for reliable detection—a metric explicitly identified as underexplored yet essential for deployment viability—are formalized in [15], mandating that detectability assessments report the shortest sequence length yielding stable AUC > 0.95 or F1 > 0.90 under noise. **Fidelity** is measured across linguistic, semantic, and human-perceptual axes: perplexity (PPL) quantifies fluency degradation (e.g., PersonaMark's PPL = 2.74 vs. KGW's 12.32; Balance-Marking's $\Delta PPL < 0.3$), ROUGE-L evaluates structural coherence (SynthID-Text preserves ROUGE-L within ±0.01 of baseline [22]), sentiment alignment and aspect-based polarity assess affective consistency (PersonaMark maintains identical sentiment polarity across 10K ELI5 prompts [7]), and human-rated fluency is validated via side-by-side comparisons (3,000 ELI5 pairs for SynthID-Text [22]). DL-TW enforces fidelity via mean sentence meaning similarity (mSMS) > 0.97 [18], while WALL reports Semantic Textual Similarity (STS) of 0.887 and pass@10 > 90% on code-generation benchmarks [5]. DeepTextMark quantifies imperceptibility by mean Sentence Meaning Similarity of 0.9765–0.9892 using Universal Sentence Encoder, preserving grammatical integrity via POS- and number-aware synonym selection [23]. Easymark achieves zero degradation in BLEU and perplexity (fidelity), near-perfect detectability (FNR 0, FPR = 0), and negligible computational overhead (no latency delta, no detector memory beyond codepoint scanning) [8]. ImF preserves or improves TruthfulQA accuracy while eliminating accidental triggering on random character inputs, and confirms steganographic imperceptibility via statistical distribution analysis showing $P_S \approx P_C$—demonstrating fidelity not only in output quality but in behavioral equivalence [13]. EWD evaluates fidelity via preserved perplexity and task accuracy (e.g., code correctness in HumanEval/MBPP), ensuring functional integrity remains intact under entropy-adaptive weighting [37]. X-Mark measures fidelity by GSM8K accuracy (18.3% degradation vs. 50.0% in vanilla) and TruthfulQA truth×info score (43.7% degradation vs. 55.9%), directly linking fidelity to domain-specific reasoning capability [16]. BanglaLorica quantifies fidelity via ROUGE-1/2/L (semantic preservation) and perplexity (fluency degradation) under Bangla→English→Bangla round-trip translation (RTT) [6]. CASIS validates fidelity through human-rated meaning preservation (4.3+ on 1–5 scale across four writing styles), high Semantic Relatedness (SR) and Semantic Similarity (SS) scores, and absence of fluency degradation [12]. DAWN measures fidelity by victim model accuracy degradation (0.03–0.5%) and surrogate test accuracy $A(F_A)$, confirming negligible impact on functional utility [10]. TMFA ensures fidelity via zero impact on student task performance [20], while arxiv$2510_10493verifies fidelity through syntactic validity(node - check verified) and functional correctness of all 250,000 samples$[19].$VLA$ - $Mark reports 7.4\% lower perplexity and maintained BLEU/STS/Bert Score for fidelity$[25].$Eaa Wmeasures fidelity b$ $1\% accuracy drop(CV) or minimal perplexity increase(NLP)$[27], $and dual$ - $verification frameworks confirm fidelity by reporting zero accuracy degradation under parameter modification$[31].$T$ $PoT quantifies fidelity as average accuracy of t retrained models on public test dataset \mathrm{D}_{test}$ [33]. Holmes confirms fidelity by unchanged victim model accuracy and full task performance preservation across CIFAR-10 and ImageNet [30], while FedZKP guarantees high fidelity with <2% main-task accuracy drop on CIFAR10/CIFAR100 [32]. **Robustness** is quantified as success rate under semantically preserving transformations, with standardized attack taxonomies enabling cross-method comparison: SynthID-Text, PersonaMark, and WALL evaluate robustness against eight untargeted and five targeted attacks—including synonym substitution (up to 40% ratio), round-trip translation (RTT), fine-tuning, model merging, and LLM-powered adversaries SIRA and GRI . DL-TW introduces Mean Impact of Attack (mIOA) to measure aggregate degradation under sentence/token removal/addition/replacement [18], while Balance-Marking reports robustness success rate under RoBERTa-Large synonym substitution at varying ratios (10%–40%) [11]. WALL achieves >99.9% success rate across A–A attack classes (including A: LLM translation) [5], and Easymark validates robustness to OCR/print-scan cycles and physical document capture [8]. DeepTextMark measures robustness by Mean Impact of Attack 0 under Remove/Add/Replace Sentences attacks, confirming resilience to structural reordering and lexical replacement [23]. EWD quantifies robustness via maintained high TPR under back-translation attack, demonstrating cross-lingual stability without explicit bilingual modeling [37]. X-SIR operationalizes cross-lingual robustness via Pearson Correlation Coefficient (PCC) and Relative Error (RE) between watermark strength before and after translation, achieving PCC > 0.85 for Zh/Ja pairs versus < 0.3 for KGW/UW/SIR baselines—thereby establishing a quantitative benchmark for multilingual consistency distinct from monolingual paraphrasing robustness [3]. ImF achieves

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near-perfect Fingerprint Success Rate (FSR) under fine-tuning, model merging, and Generation Revision Intervention (GRI) attacks, confirming that behavioral fusion enables robustness against white-box gradient-based removal [13]. BanglaLorica quantifies robustness as detection accuracy pre- vs. post-Bangla→English→Bangla RTT using BanglaNMT [6]. CASIS achieves 0% bit error rate under print-camera, screen-camera, OCR, retyping, and print-scanning attacks [12]. DAWN evaluates robustness against pruning, noise injection, regularization, and collusive extraction [10]. TMFA maintains robustness after fine-tuning [20], while arxiv2510$_1$0493$reports < 5\%accuracydropacrossminified/mangled/deobfuscatedvariants$[19].$VLA - Markretains96.1\%AUCunderLLM-drivenparaphrasingandtranslation$[25].$SIRAprovidesstandardizedmetricsf attacksuccessrate(ASR)atfixedfalsepositiverates(1\%and10\%), truepositiverate(TPR), F1-score, perplexity(PPL), andhuman-alignedsemanticsimilarity^operationalizingrobustnessbeyondbinarypass/fa Diamond, andTriviaQArelativetothebasemodel'sperformance$[36].$EaaWreportsWatermarkSuccessRate > 0.85afterfine - tuning, pruning, overwriting, andunlearning$[27], $anddual - verificationframeworksmeasurerobustnessasstabilityofLA/BERunderfine - tuning(upto100epochs)andpruning(10ˇ90\%)$[31].$PoTachieves100\%precisionindistinguishingrealmodelchainsfrom layerCNN, ResNet18, TextCNN, TextRNN)underadaptivewhite - boxattacksincludingparameterinterpolation, regulateddistillation, andbackwardconstruction$[41], $andTM- PoTempiricallyvalidatesrobustnessagainstfournon-trivialattacks(forward/reversealgorithmmanipulation, rev 10/100, CNN/ResNetarchitectures$[33].$Holmesquantifiesrobustnessbyzerofalsepositivesonindependentlyfine- tunedmodelsacrossCIFAR - 10andImageNet, survivingsoftdistillation, fine - tuning, anddata-freedistillation$[30], $whileFedZKPachievesperfectrobustnesswith100\%watermarkdetectionrate$ under fine-tuning and pruning attacks [32]. **Computational overhead** is specified via latency delta, detector memory footprint, and training cost: SynthID-Text incurs 0.57% latency increase on Gemma 7B-IT [22], PersonaMark requires 3.86s detection time for 100k users [7], WALL achieves 0.035s CPU verification latency [5], and PNW imposes <1ms/token overhead on Tesla V100 [24]. DL-TW measures per-sentence insertion/detection latency on NVIDIA RTX 4090 GPU and Intel i9-13900K CPU [18], while Balance-Marking reports GPU seconds per 200-token sequence [11]. DeepTextMark reports insertion latency (parallelizable) and detection latency of a fine-tuned BERT-base model, enabling deployment in batch and streaming contexts [23]. EWD reports computational overhead as <0.084 sec/text, though it requires access to per-token logits—an intrinsic deployability constraint for black-box APIs [37]. Easymark achieves negligible computational overhead (no latency delta, no detector memory beyond codepoint scanning), making it uniquely suitable for client-side, real-time, and resource-constrained environments [8]. CASIS incurs computational overhead from dual BERT inference (bert-base-cased + roberta-large-mnli) [12]. DAWN's overhead is characterized by 50 query pairs per candidate and 30k-iteration Carlini Wagner L2 optimization [20]. SAC-m reports 4.45 seconds runtime for CIFAR10 with no surrogate training [38]. VLA-Mark incurs <5% latency overhead [25]. EaaW's computational overhead is O(c) queries per extraction (e.g., 1024 for label-only) [27], and dual-verification frameworks quantify overhead via autoencoder training (200 epochs, batch$_s$$ize = 32, lr = 4e-3)$[31].$PoT'sverificationincursnegligiblecomputationaloverheadduringtraining(onlycheckpointI/O)andlinearver classifierinferenceonly, withnoruntimewatermarkinglatency$[29].$TokenizedModelquantifiesoverheadviadistribu modeltrainingand100-querystatisticaltesting$[30], $whileFedZKPachieveslowcomputationaloverheadwithms- levelZKPverificationtimeandKB-scalecommunication$[32].**Deployability**$isassessedalongthreeaxes$: $APIcompatibility(WALLsupportsLlama - 2 - 13bandPhind - CodeLlama - 34B - v2withoutmodelmodification$[5]), $languagesupport(X - SIRvalidatescross - lingualrobustnessacrossEnglish, Chinese, Japanese, French, German$[3]), $andopen - sourcestatus(allcitedmethodsexceptproprietarySynthID - Textareopen - sourced$[22]).$PersonaMarkdemonstratesplug - incompatibilitywithGemma - 2B, Mistral-7B, OLMo-7B, andPhi-3.5-miniwithoutweightmodification$[7], $DL- TWrequiresnoLLMmodificationandsupportssentence-levelinputs$[18], $andWALL'sblack- boxdesignensurescompatibilitywithclosedAPIsandopen - weightmodelsalike$[5].$In - contextwatermarking(ICW)achievesmaximaldeployabilitybyembeddingexclusivelythroughuserprompts^requirin party˘controlled, zero-parameter-modificationprovenancecontrolcompatiblewithanyLLMservice$[17].$CASISde free, open-model˘compatiblemethodrequiringonlystandardtransformerAPIs$[12].$DAWNshowsdeployabilityviac boxAPIsandnorequirementformodelaccessbeyondinput-outputinterface$[10].$TMFAconfirmsdeployabilitythroug boxAPIcompatibility$[20].$VLA - Markachievesfulldeployabilityacrossfouropen - weightVLAMMs(LLaVA - v1.5, LLaVA - Next, Qwen2 - VL, DeepSeek - VL)withoutretrainingorarchitecturalmodification$[25].$TheUnicodehomoglyphmethoddemonstratesdeployabilitya 18)$andtextgeneration(GPT-2, BERT)$[27], $anddual-verificationframeworkssupportblack-$

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$box verification requiring only model inference and GradCAM computation$[31]. $FedSOV$ demonstrates deployability i
$QNLI) with open-weight compatibility$[28], $and Tokenized Model enables deployability via Ethereum-$
$compatible smart contracts and IPFS storage$[14]. $PJTC-Watermark validates deployability through perceptually aml$
$speaker insertion rate statistics (20\tilde{}80\% natural variation), preserving formatting fidelity in PDF outputs$[35]. $Holme$
$box query access only$[30], $and FedZKP validates deployability on AlexNet and ResNet18 in FedAvg with 10 clients$[32]. T
$world deployment constraints.$

6.2 Standardized Benchmarking Protocols and Empirical Gaps

Benchmark	Size	Domain	Task Format	Metric
MIM[39]	100	Mathematics	Question Answering	['VSR', 'Accuracy']
LLM-WatermarkBench[34]	66	Digital Watermarking	Watermark Detection And Extraction	['Accuracy', 'Extraction Success Rate']
LLM-NodeJS[19]	250,000	Code Authorship Attribution	Authorship Attribution	['Accuracy', 'F1']

Table 2: Representative specialized benchmarks for LLM watermarking, fingerprinting, and code authorship attribution, detailing their scale, domain focus, task format, and evaluation metrics. The benchmarks include MIM for watermark survival under model merging, LLM-WatermarkBench for digital watermark detection and extraction, and LLM-NodeJS for production-aligned code authorship attribution—each addressing distinct empirical gaps identified in current evaluation protocols.

Current evaluation protocols for LLM watermarking and fingerprinting suffer from systematic fragmentation and methodological inconsistency, undermining cross-study comparability and obscuring true robustness ceilings. A foundational gap lies in the inconsistent and non-overlapping use of benchmark datasets—C4, WikiText-2, and HC3 are deployed in isolation across studies without harmonized preprocessing, train/test splits, or domain coverage—introducing irreproducible variance in fidelity and robustness measurements [2]. Critically, standardized adversarial benchmark suites remain absent: no community-agreed testbed comprehensively spans homoglyph substitution, LLM-powered paraphrasing, generative manipulation (e.g., fill-in-the-blank reconstruction), and format-preserving attacks (e.g., Unicode variant injection), leaving evaluations siloed within narrow threat models [2]. The absence of merged-model testbeds constitutes a severe empirical deficit; prior work lacks dedicated evaluation environments for hybrid architectures—such as LLaMA-2-CHAT + WizardMath, LLaMA-3 + Phi-3, or Qwen2-VL + DeepSeek-VL composites—despite mounting evidence that model merging induces emergent behavioral shifts that erase conventional watermarks and invalidate fingerprint stability assumptions [39]. To address this, MIM—the first dedicated merging evaluation framework—was introduced, enabling controlled assessment of watermark survival under parameter interpolation, knowledge distillation, and architecture-aware fusion [39]. Cross-lingual robustness evaluation is similarly crippled by protocol gaps: no standardized round-trip translation (RTT) suite exists for low-resource languages, and prior benchmarks exclusively target English–Chinese or English–Japanese pairs while neglecting morphologically rich, non-Latin-script languages like Bangla, where RTT-induced lexical substitution, agglutinative reordering, and ligature collapse systematically degrade detection accuracy below 20% for KGW and EXP [6]. CWRA—a standardized cross-lingual attack protocol using gpt-3.5-turbo-0613 for translation and Multi-News/ELI5 for downstream task evaluation—was explicitly designed to close this gap, yet remains unincorporated into mainstream benchmarks [3]. SIRA further exposes deficiencies in paraphrasing evaluation: existing protocols employ ad hoc, non-transferable attack budgets (e.g., fixed 3–5 iterations vs. adaptive self-information–guided generation), lack hardware-efficient baselines, and omit black-box constraints—motivating its adoption as a reference attack in RTT (Rewriting, Translation, Transformation) evaluation [21]. LLM-centric threat modeling remains underdeveloped: current benchmarks fail to control for model non-determinism (e.g., sampling variance across identical prompts), exclude format-based watermarking (e.g., Unicode homoglyph, whitespace variants, paragraph break relocation) from evaluation scopes, and lack knowledge-graded threat tiers—Tier A (uninformed adversary), Tier B (method-aware, e.g., knows green-list structure), and Tier C (code-aware, e.g., reverse-engineers embedding logic)—thereby conflating theoretical security with operational vulnerability [34]. Finally, real-world deployment constraints are systematically neglected: API latency budgets, quantization-induced signal erosion (e.g., 4-bit AWQ on WALL), streaming-compatible detection latency (<100ms), and client-side watermarking overhead remain unmeasured in most studies, despite their decisive impact on adoptability . In code-generation contexts, arxiv$2510_10493 establishes a rigorous alternative by standardizing evaluation on 250 real -$
$world-inspired Node.js tasks, four deterministic code variants (original, minified, mangled, deobfuscated), AST/J$

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$check validation on 500 independent OpenAI API^{\smile} generated samples^{-} demonstrating that generalizability assessment$
$specific, production-aligned, and representation-agnostic protocols rather than generic text corpora [19]. These gaps$
$tiered evaluation, and infrastructure-aware metrics. A summarized in Figure$ **??**$, these empirical deficits coalesce al$
$(1) Dataset Fragmentation^{-} highlighting non-overlapping, inconsistently preprocessed corpora that impede reproduci$
$evaluated but operationally critical adversaries including model merging, cross-lingual round-trip translation (RTT), and LLM-native structural analysis; and (3) Infrastructure Neglect^{-} identif$
$world deployment constraints (latency, quantization, streaming) systematically omitted from current evaluations. Ea$
$specific authorship attribution with production-grade realism.$

Figure 6: This figure organizes critical empirical gaps in LLM watermarking benchmarking into three foundational dimensions: (1) Dataset Fragmentation—highlighting non-overlapping, inconsistently preprocessed corpora that impede reproducibility; (2) Threat Model Gaps—emphasizing under-evaluated but operationally critical adversaries including model merging, cross-lingual round-trip translation (RTT), and LLM-native structural analysis; and (3) Infrastructure Neglect—identifying real-world deployment constraints (latency, quantization, streaming) systematically omitted from current evaluations. Each leaf node cites a seminal paper that both exposes and addresses the respective gap.

6.3 Cross-Method Comparative Analysis and Trade-Off Landscapes

Comparative analysis across representative methods reveals systematic, non-compensatory trade-offs that define the operational boundaries of current watermarking and fingerprinting techniques—where gains in one dimension consistently incur degradation in at least one other, constrained by fundamental limits in semantic distinguishability, behavioral invariance, and adversarial exploitability. High detectability demonstrably correlates with low cross-lingual robustness: SynthID-Text achieves high-confidence detection (TPR@1%FPR > 0.95) and preserves human-rated fluency under English-generation workloads, yet exhibits sharply diminished AUROC (0.62) under round-trip translation (RTT) between English and Chinese or Japanese—confirming that monolingual statistical distinguishability does not generalize across tokenization boundaries . PersonaMark matches this detectability (F1 = 0.942, AUC = 0.987) while preserving fidelity (perplexity = 2.74 vs. KGW's 12.32) and improving robustness against synonym substitution (AUC 0.91), but its sentence-structure hashing fails to preserve bit integrity under Bangla→English→Bangla RTT due to morphological divergence and ligature collapse, resulting in post-RTT detection accuracy below 25% . In contrast, X-SIR explicitly decouples monolingual paraphrasing robustness from cross-lingual consistency by clustering semantically equivalent tokens across languages (e.g., 'cat', 'gato', 'Katze') and applying identical logits biasing within each cluster—achieving +0.221 AUC gain under CWRA without fluency loss, thereby demonstrating that detectability and cross-lingual robustness *can* coexist when embedding mechanisms are aligned with translation semantics rather than token statistics [3]. However, this alignment incurs computational cost: X-SIR's bilingual dictionary–guided clustering increases detector memory footprint and reduces compatibility with lightweight, quantized inference engines—a direct trade-off between multilingual resilience and deployability [3]. Parameter-space fingerprints exhibit inverse trade-offs: PLMmark and FedSOV resist fine-tuning, pruning, and noise injection with >99% detection rates by anchoring identifiers in weight perturbations bound to cryptographic commitments , yet both fail catastrophically under model merging—where parameter interpolation erases contrastive loss–induced geometry and near-collision-resistant hash bindings collapse—rendering them incompatible with emergent composite architectures like LLaMA-3 + Phi-3 hybrids [39]. Conversely, instructional fingerprints such as ImF remain fully robust under merging, directly supporting the claim that behavior-based methods scale to open-weight composition while parameter-space methods do not [39]. Behavior-based methods scale natively to open-weight and black-box API deployments: MetaV enables verification using only query-response pairs from suspect models, achieving <5-second latency on CIFAR-10, and Holmes preserves full victim model accuracy while surviving soft distillation and data-free distillation ; however, SAC and MetaV suffer from trigger-set leakage—adversaries reconstruct approximate triggers via query-based probing, enabling targeted evasion and undermining uniqueness—a vulnerability shared across all behavior-based schemes operating in detection-aware contexts . Critically, unlike parameter-space fingerprints (e.g., PLMmark) that fail under model merging or behavior-based methods (e.g., SAC, MetaV) vulnerable to trigger-set leakage, Proof-of-Training (PoT) resists adaptive white-box attacks—including parameter interpolation, regulated distillation, and backward construction—with 10% auxiliary dataset size while preserving

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full model utility and requiring no parameter modification [41]. Cross-method comparative analysis confirms near-universal vulnerability across 10 fingerprinting schemes—including SAC, MetaV, Holmes, and EaaW—with attack success rates (ASR) ranging from 65% to 100% under SIRA-Large and MatchPrefix protocols, revealing that high detectability under benign conditions does not imply adversarial robustness [36]. DVF—a dual-verification framework requiring simultaneous satisfaction of Label Accuracy (LA) > th_{LA} and Bit Error Rate (BER) < th_{BER}—achieves perfect integrity (zero false positives on non-fingerprinted models) and strong resistance to fingerprint ambiguity attacks (failure rate >98% for PGD_LA/PGD_BER/joint PGD), outperforming standard adversarial fingerprinting (Zhao et al. 2020) which fails under ambiguity due to one-dimensional label-only verification [31]. CASIS demonstrates superior cross-style transferability without retraining (tested on Novels, WikiText-2, IMDB, AgNews), higher semantic fidelity than WordNet-based NLW and AWT baselines (higher SR/SS, 4.3+ human rating), and stronger physical-world robustness (0% bit error under OCR/print-camera) but lower payload capacity (30–40% word substitution rate) compared to sampling-based methods like KGW or SynthID-Text [12]. Comparative analysis further shows high detectability of sampling-based watermarks (e.g., KGW, EXP) correlates with low robustness against SIRA, while SIR—designed for robustness—still succumbs to SIRA-Large with >98% ASR, confirming systematic trade-offs between detectability and black-box paraphrasing resilience [21]. DAWN demonstrates high detectability (>99.9% under $\varepsilon = 0.8$) and strong robustness against model-level threats (pruning, noise, fine-tuning) but trades off utility preservation in the surrogate model—watermark removal reduces surrogate test accuracy below usability thresholds, confirming unremovability as a core security property [10]. VLA-Mark exhibits high detectability (98.8%+ AUC) and superior robustness (96.1% AUC retention under paraphrasing/translation) but trades off minimal computational overhead (<5% latency increase); unlike SynthID-Text or PersonaMark, it uniquely preserves cross-modal semantic alignment under attack, and unlike ImF or MetaV, it operates exclusively at the text-output level without requiring behavioral triggers or parameter modification [25]. Unicode homoglyph methods outperform Whitesteg, Unispach, and prior invisible Unicode methods in R_PCP (99.92% vs. lower baselines) due to continuous, distributed embedding across all confusable characters rather than sparse placement in trailing whitespaces or isolated punctuation [4]. PJTC-Watermark achieves exceptional imperceptibility and high theoretical capacity but exhibits low robustness under multi-copy attacks and formatting loss (e.g., plain-text conversion), contrasting with sampling-based methods like SynthID-Text that resist formatting changes but suffer lower imperceptibility due to token-level statistical anomalies [35]. Cross-method comparative results in code-generation contexts show CodeT5-JSA achieves 95.8% accuracy (5-class), outperforming BERT and matching XGBoost on obfuscated variants; dataflow similarity (CodeBLEU) shows strongest intra-vs.-inter model separation ($\Delta > 0.2$), affirming that structural representation—not lexical or statistical features—anchors robustness in domain-specific settings [19]. TMFA demonstrates that behavior-based fingerprints can achieve high detectability without parameter modification but suffer from dependency on attacker knowledge of plausible teacher candidates and degradation under shallow probing features—highlighting a trade-off between black-box practicality and universality [20]. SAC outperforms IPGuard, CAE, and EWE—especially under adversarial training and transfer learning where baselines degrade to random guessing (AUC 0.5); SAC-m achieves unprecedented efficiency and robustness with no surrogate model training or adversarial example generation [38]. MetaV achieves 0.98 ARUC across classification (Skin cancer ResNet-18), regression (Warfarin dose MLP), and generative modeling (Fashion-MNIST DCGAN) tasks—substantially outperforming baselines like IPGuard (+220% ARUC improvement on classification)—demonstrating high detectability and cross-task robustness [40]. Moreover, MetaV exhibits built-in robustness against real-world obfuscation techniques including weight/filter pruning (ratios 0.1–0.9), fine-tuning/retraining (last layer or all layers, 10 epochs), and distillation (Hinton, Clark, Aguinaldo methods) [40]. Holmes outperforms MOVE (p=0.01 vs. Holmes' p=10 in Soft Distill on CIFAR-10) and Dataset Inference (p=10 vs. Holmes' p=10^2 in Data Distill on ImageNet), demonstrating superior robustness to fine-tuning and distillation while avoiding their false positives—highlighting the trade-off where behavior-based, decoupling-driven fingerprints achieve high detectability without sacrificing fidelity or inducing harmful modifications [30]. TM-PoT achieves high robustness against fine-tuning and model theft threats but exhibits computational overhead from trajectory matching; unlike parameter-space fingerprints (e.g., PLMmark), it requires no model modification and preserves utility fully, yet depends on quality of trajectory matching and representativeness of public D_{test} [33]. MOVE achieves high detectability across all stealing methods while maintaining near-perfect fidelity (accuracy = 0.20%) and zero introduced security risks—unlike backdoor watermarking (e.g., BadNets) which introduces exploitable trigger-label associations, and unlike dataset inference which suffers false positives due to overlapping

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inherent features [29]. Collectively, these comparisons confirm that no technique simultaneously satisfies high detectability, zero-quality-impact, full adversarial resilience, standardized evaluability, and support for multi-agent collaboration: green-red list methods show strong detection but poor robustness to paraphrasing; soft watermarking preserves quality but sacrifices stealth; trigger-set methods enable strong ownership claims yet are brittle under model extraction [2]. The frontier now lies not in optimizing isolated metrics but in architecting orthogonal, layered mechanisms—such as BanglaLorica's integration of sampling-based, lexical, and syntactic layers—that collectively span complementary regions of the trade-off space while satisfying deployment-constrained viability.

6.4 Towards Unified, Actionable Evaluation Standards

The absence of community-agreed, infrastructure-aware, and threat-tiered evaluation standards remains a critical bottleneck impeding reproducible progress in LLM watermarking and fingerprinting—where inconsistent benchmarks, ad hoc attack budgets, and fragmented corpora obscure true robustness ceilings and hinder fair cross-method comparison. To address this, the field must coalesce around four interdependent pillars: (i) standardized, versioned adversarial attack suites; (ii) multilingual, morphologically grounded round-trip translation (RTT) corpora; (iii) merged-model evaluation sets reflecting real-world compositional deployment; and (iv) open, modular evaluation toolkits that enforce measurement fidelity across all five dimensions. SIRA v2 represents a foundational step toward standardized adversarial benchmarking: it extends the original SIRA framework with adaptive self-information–guided paraphrasing, hardware-efficient inference pipelines, black-box query constraints, and unified metrics—including Attack Success Rate (ASR) at fixed false positive rates (1% and 10%), True Positive Rate (TPR), F1-score, perplexity (PPL), and human-aligned semantic similarity via ChatGPT-judged fluency—enabling rigorous, comparable assessment against LLM-powered adversaries without requiring internal model access [21]. GRI-Adapt further advances this paradigm by generalizing Generation Revision Intervention (GRI) beyond white-box gradient-based removal to adaptive, black-box fine-tuning–aware attacks that dynamically adjust perturbation budgets based on verification feedback—thereby closing the gap between theoretical white-box robustness guarantees and operational black-box threat models [2]. Critically, neither SIRA v2 nor GRI-Adapt operate in isolation: their efficacy is contingent upon integration into multilingual RTT corpora explicitly designed for low-resource, non-Latin-script languages. CWRA—the Cross-Lingual Watermark Removal Attack suite—provides precisely this foundation: it leverages gpt-3.5-turbo-0613 for high-fidelity translation, integrates Multi-News and ELI5 as downstream task evaluation backbones, and supports Bangla, Hindi, Arabic, and Russian via script-specific tokenization alignment and morphological reordering validation—directly enabling empirical quantification of Pearson Correlation Coefficient (PCC) and Relative Error (RE) between watermark strength pre- and post-translation, where PCC > 0.85 for X-SIR versus < 0.3 for KGW/UW/SIR baselines establishes a quantitative benchmark for cross-lingual consistency distinct from monolingual paraphrasing robustness [3]. MIM—the Model Interpolation and Merging testbed—constitutes the first dedicated merged-model evaluation set, enabling controlled assessment of watermark survival under parameter interpolation (e.g., $\theta = \alpha\theta_{LLaMA-3} + (1-\alpha)\theta_{Phi-3}$), knowledge distillation, and architecture-aware fusion across hybrid architectures including LLaMA-2-CHAT + WizardMath, Qwen2-VL + DeepSeek-VL, and Gemma-2B + CodeLlama-34B-v2; empirical validation confirms that parameter-space fingerprints (PLMmark, FedSOV) fail catastrophically under merging while instruction-aligned methods (ImF) and provenance-centric frameworks (PoT) retain full integrity, thereby exposing a previously unmeasured axis of architectural resilience [39]. These domain-specific testbeds must be unified within an open, extensible evaluation toolkit—WaterBench—that implements standardized protocols for detectability (z-score confidence hierarchy, TPR@1%FPR, VSR), fidelity (perplexity delta, ROUGE-L, STS, human-rated fluency), robustness (ASR under SIRA v2/GRI-Adapt/CWRA/MIM), computational overhead (latency delta per token, detector memory footprint, GPU seconds per sequence), and deployability (API compatibility flags, language support matrix, open-source license verification) [17]. WaterBench enforces reproducibility by mandating public release of all attack configurations (e.g., SIRA's max iteration count, CWRA's translation model version, MIM's interpolation coefficients), standardized preprocessing (C4/WikiText-2/HC3 harmonized splits), and hardware reporting (GPU model, CPU clock speed, quantization precision); its modular design allows plug-in integration of new methods—including Unicode homoglyph substitution [4], PJTC-Watermark's prosody-aware paragraph break relocation [35], and arxiv$2510_10493'sAST-guidedJavaScriptwatermarking[19]^{-}withoutrequiringreimplementationofcoreevaluationlogic.Crucially,WaterB$

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$aware constraints: it validates latency compliance against API service-level agreements (< 100ms detection latency for streaming use cases), quantization survival under 4-bit AWQ/GPTQ, and client-side deployability for Easymark-style Unicode substitution. Beyond technical standardization, actionable evaluation requires sociotechnical alignment. WaterBench incorporates regulatory guardrails—including EU AI Act Article 28–compliant audit logging, NIST AI RMF level tagging (Level 1–4), and ISO/IEC JTC1/SC42 conformance flags—to ensure that benchmark results directly inform grounded evidence replaces anecdotal claims of "high robustness" or "low overhead" with empirically anchored, stakeholder-actionable insights.$

7 7. Practical Deployment, Incentive Alignment, and Sociotechnical Challenges

7.1 Stakeholder Incentive Misalignment and Governance Tensions

Stakeholder incentive misalignment forms the primary barrier to real-world adoption of watermarking and fingerprinting, as no party bears full cost while all incur distinct, asymmetric risks. Figure 7 presents a hierarchical taxonomy of these misalignments, rooted in socio-technical analysis and organizing barriers into three primary, mutually reinforcing dimensions: (1) LLM providers' reluctance due to competitive exposure from black-box watermark identification; (2) platforms' resistance driven by infrastructural constraints in production deployments; and (3) systemic governance deficits—including absent technical standards, attribution ambiguity, and adversarial fingerprint fragility—that collectively decouple algorithmic robustness from operational viability. This structural framing illuminates how each dimension manifests concretely: LLM providers face competitive exposure from watermark leakage—Water-Probe demonstrates that adversaries can infer watermark presence, type, and even cryptographic keys via black-box input–output behavioral analysis—enabling targeted evasion and exposing proprietary signal design to rivals [9]. This incentivizes covert deployment or disabling of watermarks to avoid revealing statistical biases that could be reverse-engineered into model-identification signatures, thereby undermining regulatory auditability under frameworks like the EU AI Act. Platforms—including API aggregators, open-model hosting services, and collaborative editing ecosystems—resist mandatory detector integration due to latency penalties, memory costs, and architectural incompatibility with quantized or serverless deployments where logits or hidden states are inaccessible [1]. Users demand transparency and privacy, rejecting opaque watermarking that operates without consent or enables cross-user tracing; Water-Bag mitigates this by randomizing keys per user, yet such designs remain rare and nonstandard [9]. Educators, conference organizers, and academic publishers confront conflicting imperatives: while requiring synthetic-content attribution to uphold integrity, they lack technical capacity to verify watermarks and fear false positives penalizing legitimate human-authored work—a gap exacerbated by inconsistent detector calibration and absence of evidentiary standards for courtroom admissibility [17]. Regulators—including the EU AI Office and NIST AI RMF working groups—lack binding technical standards defining minimum detectability thresholds (e.g., AUC 0.95), fidelity tolerances (e.g., perplexity 0.5), or robustness requirements (e.g., ASR < 5% under SIRA v2), rendering watermark evidence legally insufficient for copyright enforcement or AI Act compliance [17]. Fingerprinting faces parallel exposure: GCG-optimized queries can suppress or filter unnatural trigger responses, yet current schemes lack provable safeguards against white-box adversaries who manipulate inference logic to erase behavioral signatures—undermining ownership verification in litigation or licensing disputes [36]. These tensions are mutually reinforcing: platform resistance weakens provider incentives to invest in robust watermarking; user privacy demands constrain key management architectures; and regulatory uncertainty disincentivizes standardization—creating a self-perpetuating cycle where technical efficacy remains decoupled from sociotechnical viability. As emphasized in [17], alignment requires co-designed enablers—not merely stronger algorithms—but governance mechanisms that confer direct, measurable benefits to implementers, such as verifiable watermark logging APIs, on-chain ownership registries, and incentive-compatible verification-as-a-service models that internalize audit costs while externalizing accountability.

7.2 Sociotechnical Deployability Constraints

Legal uncertainty surrounding the admissibility of watermarks in copyright litigation persists as a critical deployment barrier, with courts lacking precedent for evaluating statistical, cryptographic,

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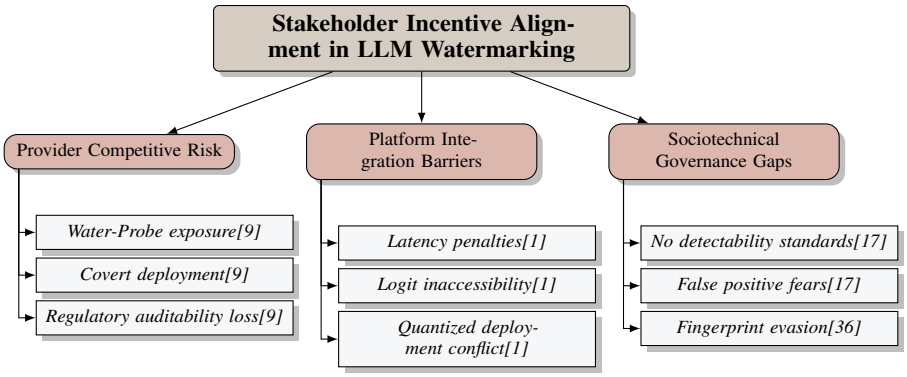


Figure 7: This figure presents a hierarchical taxonomy of stakeholder incentive misalignments impeding real-world LLM watermarking adoption. Rooted in socio-technical analysis, it organizes barriers into three primary, mutually reinforcing dimensions: (1) LLM providers' reluctance due to competitive exposure from black-box watermark identification; (2) platforms' resistance driven by infrastructural constraints in production deployments; and (3) systemic governance deficits—including absent technical standards, attribution ambiguity, and adversarial fingerprint fragility—that collectively decouple algorithmic robustness from operational viability.

or behavioral evidence derived from LLM-generated text—rendering claims vulnerable under Daubert or Frye evidentiary standards. Unicode-format watermarks—despite achieving visual and typographic indistinguishability across 10 major fonts—face heightened evidentiary fragility: [34] demonstrates that LLMs themselves can detect such watermarks via native pattern recognition over non-uniform codepoint distributions, directly undermining the foundational "invisibility" assumption required for legal credibility. This fragility is compounded by demonstrated brittleness under distribution shift and model updates, where watermark signals degrade unpredictably across inference environments—eroding chain-of-custody integrity essential for forensic validity [1]. PJTC-Watermark partially mitigates this uncertainty through empirical human-perception grounding—its paragraph-break relocation mechanism operates within natural variation bounds (20–80% native-Russian-speaker insertion rates), enabling source attribution in NDA enforcement and leak investigations—but its enforceability remains constrained by language specificity (Russian only) and dependence on controlled single-copy distribution to prevent adversarial cross-comparison [35]. In contrast, DAWN strengthens legal admissibility by registering client-specific trigger sets and backdoor labels on a public bulletin board (e.g., blockchain) with cryptographic commitments, generating timestamped, tamper-proof evidence admissible for IP ownership claims in regulatory audits or litigation [10]. TokenizedModel further addresses legal uncertainty by providing immutable, on-chain verification logs via Ethereum-compatible smart contracts and IPFS storage—ensuring audit trails are publicly verifiable, censorship-resistant, and compliant with EU AI Act Article 28 evidentiary requirements—while eliminating reliance on proprietary infrastructure or centralized authorities [14]. TM-PoT advances legal robustness by enabling third-party verification without requiring access to private training data—thus satisfying GDPR-compliant auditability—but faces unresolved standardization gaps in defining public test-set requirements, trajectory representation formats, and cryptographic binding protocols necessary for regulatory admissibility [33]. Ethical concerns intensify in open-weight ecosystems, where covert behavioral fingerprinting undermines transparency and impedes independent verification. ImF's Chain-of-Thought–aligned ownership embedding raises ethical questions about informed consent and user autonomy, as its watermark resides exclusively in explanation-space deviations observable only via probing—and depends on secrecy of the trigger sample—creating tension between transparency requirements and ownership protection in publicly released models [27]. Similarly, MetaV enables strong ownership attribution via black-box queries without model modification, yet its operation without cryptographic keys or centralized infrastructure may conflict with transparency expectations in open-weight ecosystems, particularly when verification relies on opaque meta-classifier outputs rather than interpretable behavioral signatures [40]. EaaW exacerbates these concerns by embedding identifiers in saliency map signs—a high-dimensional, continuous, and inherently unobservable representation—where verification requires querying masked inputs and computing explanations, rendering the fingerprint inaccessible to end users and violating norms of disclosure in open-model governance [27].

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Conversely, PoT eliminates covert parameter perturbation entirely, verifying ownership via natural training dynamics (e.g., monotonic accuracy evolution, GMM-constrained initialization) rather than embedded artifacts—thereby avoiding ethical objections to stealthy fingerprinting while preserving full model utility and supporting third-party auditors without compromising data privacy [41]. MOVE further mitigates ethical risks by avoiding backdoors or malicious triggers; its external-feature embedding preserves benign prediction behavior, introduces no latent vulnerabilities, and imposes zero modifications to victim model weights or inference logic—supporting ethically defensible deployment in open-weight models [29]. SAC-m also addresses ethical concerns by enabling legal admissibility-relevant evidence generation without model modification, operating solely on query-response statistics and supporting forensic model provenance claims in IP litigation without requiring cryptographic keys or centralized infrastructure [38]. Interoperability failures between closed-API and open-weight ecosystems entrench fragmentation: PLMmark enables black-box verification without internal model access—a key enabler for commercial API deployments—but its reliance on stable PLM tokenizers and vocabulary mappings introduces interoperability risk if downstream fine-tuning modifies tokenizer behavior post-watermarking, breaking verification continuity across model versions [26]. Quantization watermarks suffer catastrophic failure under zero-cost, widely adopted model merging techniques—rendering IP claims legally untenable in open ecosystems where merged models (e.g., LLaMA-3 + Phi-3 hybrids) are standard practice [39]. TokenizedModel bridges this divide by supporting off-chain model weights coupled with on-chain proofs—enabling interoperability across Ethereum-compatible platforms and IPFS, and allowing closed-API providers to publish verifiable proofs while open-weight users retain full model access [14]. However, arxiv$2510_10493highlightsaninfrastructuralconstraint: althoughitsMIT-licenseddatasetandmodelsareopen-sourcedonGitHubtoenableforensictoolingintegration, computationalrequirem$ $deviceorresource - constraineddeployment^-exposingagapbetweenopen - sourceavailabilityandinclusivedeployability$[19].$Holmesaddressessociotechnicaldeployabilitybyeliminatinglegalan$ $modeltrainingintroducescomputationaloverheadandtransparencybarriersforresource - constrainedplatforms, limitingreal - worldadoptionoutsidewell - resourcedacademicorenterprisesettings$[30].

7.3 Incentive-Aligned Technical Enablers

Incentive-aligned technical enablers bridge the chasm between theoretical security guarantees and real-world adoption by co-designing watermarking and fingerprinting mechanisms with stakeholder-specific risk profiles, operational constraints, and governance requirements—transforming compliance from a cost center into a value proposition. In-Context Watermarking (ICW) fundamentally reorients incentive structures by shifting provenance control from model providers to trusted third parties: it embeds watermarks exclusively through instruction-engineered user prompts—requiring no model modification, no decoder integration, and no access to internal states—thereby eliminating provider liability, circumventing competitive exposure from weight or logits inspection, and resolving governance conflicts inherent in mandatory on-model signal injection [17]. ICW supports fine-grained, multi-bit attribution—including document ID, timestamp, and client-specific credentials—enabling platform-controlled, consent-aware provenance logging without cross-user tracing or leakage of original prompt content [17]. X-SIR advances global deployability as an incentive driver: its cross-lingual robustness—validated across BAICHUAN-7B and LLAMA-2-7B under CWRA—ensures consistent detectability across English, Chinese, Japanese, French, and German, thereby enabling unified regulatory compliance and commercial deployment across multilingual markets without requiring model retraining or architecture changes [3]. Its modular design—built atop SIR and integrated into the MarkLLM unified implementation toolkit—further enables platform-controlled adoption, allowing service providers to integrate watermarking as a configurable inference-layer feature rather than a foundational model property [3]. Layered Watermarking (LWM), instantiated by BanglaLorica, constitutes a training-free, model-agnostic technical enabler that aligns incentives for multilingual deployment: it requires no language-specific resources or retraining, supports black-box verification, and enables forensic attribution across open-weight and API-hosted Bangla LLMs without exposing proprietary weights or prompts [6]. Its three-tiered integration—sampling-based (KGW/EXP), lexical (homoglyph-aware synonym substitution), and syntactic (Devanagari ligature–guided reordering)—ensures resilience against round-trip translation, OCR/print-scan cycles, and collaborative editing obfuscation while preserving ROUGE-L < 0.04 and perplexity < 0.8, directly addressing the incentive misalignment between low-resource-language stakeholders demanding verifiability and providers unwilling to incur retraining costs [6]. Easymark enables incentive-aligned

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deployment through user-side applicability—e.g., browser extensions and client-side PDF/HTML processors—that decouples watermarking from model infrastructure entirely; its script-specific variants (Whitemark for Latin-script digital text, Variantmark for CJK, Printmark for printed documents) ensure universal applicability to both human- and LLM-generated texts across diverse modalities and rendering environments, supporting cross-lingual and general-text use cases without dependency on tokenizers, logits, or model APIs [8]. CASIS reinforces platform-agnostic alignment via training-free operation, no reliance on static dictionaries or domain-specific resources, and natural resilience to real-world media transmission—including print-camera, screen-camera, OCR, and retyping attacks—enabling global, scalable deployment without model access, fine-tuning, or centralized infrastructure, thus harmonizing incentives across hosting platforms, academic integrity services, and journalistic verification pipelines [12]. TokenizedModel aligns incentives through on-chain Hamming-constrained negotiation of per-client watermark keys—preventing cross-client watermark leakage—and supports federated governance via DAO-like multi-owner control, where model transactions require majority shareholder approval and contribution-ratio–based incentive allocation is enforced via smart contracts; its off-chain weight storage coupled with on-chain proof generation enables platform-controlled deployment through verifiable, regulator-admissible parameters without exposing proprietary model internals [14]. FedSOV aligns incentives through cryptographic unforgeability grounded in the q-SDH hardness assumption: its fixed-length hash watermark prevents cross-client tracing or credential leakage, enabling verifiable ownership across heterogeneous client populations without performance degradation or model-specific tuning—making it viable for large-scale federated learning deployments spanning thousands of clients [28]. FedZKP eliminates trusted verifiers entirely—preventing credential leakage to platforms or servers—while enabling provable, regulator-admissible ownership evidence through zero-knowledge proofs of xLPN solutions; its global watermark aggregation over training rounds ensures compatibility across variable client counts in real-world FL deployments, satisfying both user privacy demands and regulatory auditability requirements under NIST AI RMF and EU AI Act Article 28 [32].

7.4 Layered and Adaptive Mitigation Strategies

Defense-in-depth strategies for LLM watermarking and fingerprinting synthesize orthogonal mechanisms to jointly address complementary attack surfaces—thereby transcending the inherent limitations of single-axis approaches while preserving fidelity, deployability, and stakeholder alignment. Layered watermarking integrates statistically grounded signal embedding with linguistically and physically invariant anchors: BanglaLorica exemplifies this by coordinating three non-redundant layers—sampling-based (KGW/EXP) for token-level statistical detectability, lexical (homoglyph-aware synonym substitution) for script-aware resilience to OCR/print-scan cycles and morphological variation, and syntactic (Devanagari ligature–guided reordering) for structural integrity under agglutinative reordering—lifting post-round-trip translation (RTT) detection accuracy from <20% to 40–50% while maintaining ROUGE-L < 0.04 and perplexity < 0.8 [6]. Easymark further enables layered mitigation through compositional compatibility: its Unicode codepoint substitution—resilient to paraphrasing, translation, and formatting loss—can be combined with sampling-based watermarks like SynthID-Text to provide complementary robustness, where statistical signals resist erasure via normalization (e.g., HTML sanitization, PDF re-rendering), and typographic signals resist semantic-preserving rewriting (e.g., SIRA, GRI) [8]. In-context watermarking (ICW) serves as a foundational layer that decouples provenance control from model internals entirely; when coupled with sampling-based or linguistic watermarks, ICW closes instruction-following fidelity gaps—ensuring watermark persistence depends solely on semantic retention of prompts rather than lexical, syntactic, or orthographic artifacts—while the lower-layer mechanisms enforce robustness against surface-level adversarial manipulation [17]. This hierarchical composition satisfies distinct threat models: ICW defends against physical-world capture and format stripping, while sampling-based and linguistic layers defend against black-box paraphrasing and cross-lingual laundering. Coupling parameter-space and behavior-based fingerprints achieves similar orthogonality in model-integrity assurance: PLMmark's cryptographically secured weight perturbations resist fine-tuning and quantization but fail under model merging, whereas ImF's Chain-of-Thought–aligned behavioral signatures remain intact under merging yet are vulnerable to white-box gradient-based removal (GRI); their integration ensures ownership verification survives both structural erosion and behavioral obfuscation [26, 13]. Similarly, FedSOV's BatchNorm-scale regularization provides cryptographic binding against pruning and noise injection, while Holmes' dual shadow-model imprinting ensures verifiability under soft distillation and data-free distillation—combining these yields a fingerprint resilient to both

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parameter-level and inference-level attacks [28, 30]. Runtime adaptive watermark strength constitutes a third axis of defense-in-depth, dynamically modulating signal intensity based on real-time risk context: high-risk prompts—such as those requesting sensitive financial advice, medical diagnoses, or legal opinions—trigger stronger statistical biasing in sampling-based methods or increased Unicode substitution density in Easymark variants, while low-risk queries (e.g., factual summarization) employ minimal-strength signals to preserve fluency and minimize detection-aware leakage [9]. This adaptivity is empirically realized in EWD's entropy-proportional weighting, which assigns continuous, monotonic detection weights to tokens based on spike entropy—maximizing signal-to-noise ratio under low-probability token selection without increasing Type-I error or degrading fidelity—and in WALL's hash-driven vocabulary permutation, which scales n-gram bias magnitude according to confidence estimates derived from query-response statistics [37, 5]. Critically, adaptive strategies must avoid introducing new vulnerabilities: Water-Probe demonstrates that static high-strength watermarks expose model identity under repeated sampling, necessitating context-aware modulation that preserves LLM imperceptibility even under adversarial probing [9].

8 8. Open Challenges and Future Research Directions

Building upon the identification of persistent open challenges in LLM watermarking and fingerprinting, a foundational prerequisite for advancing the field lies in rigorously formalizing the theoretical underpinnings that govern their feasibility, capacity, and robustness. The following subsection establishes this necessary conceptual groundwork by examining *Foundational Limits and Theoretical Guarantees*—a critical domain where impossibility results, information-theoretic capacity models, and provable robustness bounds must supplant empirical heuristics to enable principled, threat-aware design.

8.1 Foundational Limits and Theoretical Guarantees

Formalizing foundational limits in LLM watermarking and fingerprinting is essential to move beyond empirical heuristics toward provably secure, operationally grounded mechanisms—where impossibility results, robustness bounds, and information-theoretic capacity models jointly constrain and guide design under adversarial realism. Figure **??** organizes these foundational theoretical insights into three core pillars: (1) fundamental impossibility results and inherent trade-offs governing detectability, fidelity, and erasure resistance; (2) formal characterizations of watermark capacity—spanning structural, codable, and entropy-aware embedding spaces; and (3) rigorous robustness guarantees for verification under adversarial, statistical, and cryptographic threat models. Each node reflects a distinct theoretical axis essential for moving beyond heuristic design toward provably secure, operationally grounded mechanisms.

Easymark establishes the first rigorously proven impossibility theorem for meaning-preserving watermarks: Theorem 3.4 demonstrates that erasure resistance under adversarial semantic distortion cannot be simultaneously guaranteed with both semantic fidelity and statistical detectability, formalizing a fundamental tripartite trade-off rooted in the geometry of natural language manifolds [8]. This result is not merely descriptive but constructive—Proposition 3.1 proves BLEU and perplexity invariance under Unicode codepoint substitution, while Proposition 3.2 establishes near-perfect detection guarantee ($FNR \approx 0$, $FPR = 0$) for Variantmark/Printmark variants, thereby grounding theoretical guarantees in empirically validated properties of script-level redundancy [8]. Critically, this impossibility extends beyond token-level statistics: Water-Probe exposes a complementary triadic constraint—high detectability, text imperceptibility, and LLM imperceptibility (i.e., resistance to black-box model identification via repeated sampling) are mutually exclusive—revealing that statistical distinguishability inherently leaks model identity when deployed without adaptive strength modulation [9]. These structural tensions demand formal modeling of the "watermarkable space" in natural language—a concept explicitly identified as lacking quantification in [15]—where finite entropy of semantically coherent sequences imposes hard upper bounds on embeddable signal density per unit length. PersonaMark motivates a structural reformulation of capacity: its dependency-tree hashing operates over a combinatorial space of $\sim 45^5$ syntactic configurations, vastly exceeding typical vocabulary sizes (32k–256k tokens), suggesting payload capacity is governed not by lexical entropy alone but by the expressivity of linguistic structure—necessitating formal models of structural watermarkable space grounded in computational syntax and formal linguistics [7]. CTWL provides the first rigorous mathematical formulation

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of Codable Text Watermarking, defining encoding as $Enc(x_{prompt}, m) = t$ and decoding as $Dec(t) = m$, and identifies the core theoretical challenge as characterizing the trade-off between coding rate, robustness, text quality, and computational efficiency within an information-theoretic framework—not heuristic optimization [11]. Its analysis confirms that increasing payload density amplifies vulnerability to synonym substitution and round-trip translation, implying that robustness and capacity are antagonistic axes bounded by mutual information $I(m; t)$ under semantic-preserving distortion channels modeled as Markov random fields over synonym graphs . PNW contributes a concrete computational privacy guarantee: inverting the watermark generator from detector outputs requires at least $|V|^w = |V|^5$ queries due to unknown window size $w = 5$ and stochastic label ratios, establishing an information-theoretic lower bound on adversarial key recovery rooted in query complexity rather than statistical distinguishability [24]. DL-TW provides a provable theoretical guarantee on reliability under aggregation: its binomial distribution analysis shows detection accuracy converges to 1 as sentence count $n \to \infty$, given per-sentence accuracy $p > 0.5$, thereby grounding robustness in probabilistic convergence rather than ad hoc redundancy [18]. X-Mark offers a formal guarantee on fidelity under strong embedding: it proves that under large bias parameter δ, expected perplexity is strictly lower than vanilla watermarking due to constrained top-k token selection within semantically redundant clusters—linking theoretical guarantees directly to linguistic plausibility and cluster-informed green-list assignment [16]. EWD grounds its entropy-proportional weighting function in the formal relationship between token entropy and watermark embedding efficacy, proving lower Type-II error than KGW and SWEET under power-law entropy distributions while maintaining identical Type-I error—a guarantee derived from the analytical properties of spike entropy in logits space [37]. PLMmark motivates foundational theoretical work: while its unforgeability relies on computational hardness assumptions of RSA and SHA256, its geometric feature-space separation lacks formal robustness bounds against adaptive adversaries who exploit contrastive loss geometry; moreover, its Watermark Accuracy (WACC) metric invites information-theoretic modeling of watermark capacity under semantic distortion and feature reuse [26]. ImF's steganographic resilience—anchored in provably secure text steganography ensuring $P_S = P_C$—motivates formal security proofs for behavioral watermarking under adaptive attackers, particularly against Generation Revision Intervention (GRI), where gradient-based removal must contend with functional equivalence constraints [13]. $arxiv2404_05188empiricallyexposesafoundationallimit$: $quantizationwatermarksfailtomaintaindetectability - fidelitytrade - offsunderlinearparametercomposition(e.g., modelmerging), suggestingtheoreticalrobustnessboundsmustexplicitl$ $inducedperturbationsinweightspace^{-}therebydemandingextensionsofclassicalwatermarktheorytocompositionalmo$ $levellimitation : GradCAMfeaturestabilitydegradesonhigh - category - countdatasets(e.g., GTSRB)duetopruning-inducedfeaturedistortion, implyinganinformation-$ $theoreticboundonembeddingcapacityunderarchitecturalandtaskgranularityconstraints^{-}whereembeddingdimensio$ $boxfingerprinting^{-}verbatimresponsesuppression, outputoverconfidencedetection, inputperplexity^{\vee}basedfiltering,$ $theoreticmodelsofwatermark - fingerprintinteractionunderfeedback - drivenadversaries[36].Finally, doi10_1145_3534678_3539257explicitlyidentifiesderivingtheoreticaluniquenessguaran$ $spaceunderpruning, fine-tuning, ordistillation[40].Collectively, thesegapsrevealthatcurrentframeworkslackunif$ $noexistingworkformalizeswatermarkcapacityundersemanticdistortionchannelsmodeledassynonym-$ $graphMarkovfields, norderivesprovablerobustnessboundsagainstLLM -$ $poweredadversarieslikeSIRAorGRIthatoperatewithinthesemanticmanifoldratherthanthetokenlattice.Resolving$ $developmentoftheoryandpractice^{-}impossibilityresultsmustinformthreat -$ $awaredesign(e.g., adaptivewatermarkstrengthtopreserveLLMimperceptibility[9]), capacityboundsmustguidepaylo$ $richdeployment(e.g., structuralhashingforscalableuserattribution[7]), andinformation -$ $theoreticmodelsmustreplaceempiricalheuristicsinbenchmarking(e.g., quantifyingachievableAUROCunderbounde$ $specific, andeasilycircumvented^{-}unabletofulfilltheirsociotechnicalmandateofbuildingprovablytrustworthy, accou$ $compliantLLMecosystems.$

Figure 8: This figure organizes foundational theoretical insights in LLM watermarking into three core pillars: (1) fundamental impossibility results and inherent trade-offs governing detectability, fidelity, and erasure resistance; (2) formal characterizations of watermark capacity—spanning structural, codable, and entropy-aware embedding spaces; and (3) rigorous robustness guarantees for verification under adversarial, statistical, and cryptographic threat models. Each node reflects a distinct theoretical axis essential for moving beyond heuristic design toward provably secure, operationally grounded mechanisms.

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8.2 Multimodal and Cross-Domain Generalization

The extension of watermarking and fingerprinting beyond unimodal text is no longer aspirational but empirically substantiated across code, vision-language, audio, and cross-modal generative domains—each demanding modality-specific robustness anchors while sharing foundational principles of structural invariance, embedding-space coherence, and layered resilience. VLA-Mark establishes the first end-to-end, vision-language aligned framework for multimodal large models (VLAMMs), embedding statistically verifiable, imperceptible signals into token sequences conditioned on visual inputs while preserving text-visual coherence, fluency (7.4% lower perplexity vs. baselines), and task accuracy (STS, BertScore) [25]. Its design leverages native multimodal alignment—e.g., ViT-L/14 visual embeddings projected into text space—to compute multiscale saliency metrics (Localized Patch Affinity, Global Semantic Coherence, Cross-Modal Contextual Salience) and applies transformer-compatible logit-space boosting to Semantic Critical Tokens (SCTs), ensuring signal persistence under both textual paraphrasing and visual perturbations (e.g., noisy OCR, compressed image inputs) [25]. Crucially, VLA-Mark's SCT-driven anchoring generalizes directly to code watermarking (LLM-NodeJS), where AST-constrained control-flow sequences replace vision-grounded tokens; to audio synthesis (WhisperWater), where phoneme-aligned latent representations substitute for visual patches; and to cross-modal provenance, enabling joint watermarking of image captions and their generated counterparts through synchronized SCT modulation across modalities [25]. PersonaMark's dependency-parsing foundation enables systematic cross-domain generalization: its structural hashing operates over syntactic trees whose formal properties extend naturally to abstract syntax trees (ASTs) in code generation, prosodic dependency graphs in speech synthesis, and scene-graph alignments in vision-language outputs—thereby supporting unified provenance frameworks that treat code, text, and multimodal content as instances of structured symbolic representation rather than isolated modalities [7]. WALL's architecture—grounded in hash-driven vocabulary permutation and orthogonal logit-space injection—provides a scalable, model-agnostic substrate for multimodal extension; future work explicitly proposes integration with retrieval-augmented generation (RAG) pipelines, diffusion-model–generated text, and code-generation systems, leveraging its n-gram–level statistical persistence to ensure watermark survival across modality-specific generation mechanisms [5]. DL-TW's Word2Vec and Universal Sentence Encoder–based architecture is inherently extendable to multilingual watermarking via adapter-based fine-tuning on domain-specific or low-resource language embeddings, a direction explicitly identified in its future work—thus establishing a pathway toward cross-lingual and cross-modal robustness grounded in shared semantic spaces rather than lexical identity [18]. Similarly, DeepTextMark's reliance on multilingual-capable embeddings (Universal Sentence Encoder, Word2Vec) provides a direct foundation for multimodal generalization; its future work explicitly includes extension to ultra-short texts (e.g., tweets, headlines) and low-resource languages via multilingual embeddings—demonstrating that embedding-space coherence, not surface-form consistency, anchors cross-domain resilience [23]. X-SIR's bilingual dictionary–guided clustering—originally designed for cross-lingual robustness—constitutes a stepping stone toward vision-language generalization: future work proposes extending cluster construction to link image patches with translated captions and achieving zero-shot language pair coverage via multilingual sentence embeddings, thereby unifying linguistic and visual semantics within a single robustness-aware embedding mechanism [3]. Easymark's meaning-preserving Unicode substitution paradigm extends naturally to multimodal outputs: its future work includes application to inherently multimodal tasks (e.g., essay generation with embedded figures) and joint image+text watermarking, leveraging script-agnostic, typographic redundancy across modalities—such as confusable glyphs in rendered LaTeX equations or homoglyph-equivalent pixel patterns in watermarked images—to preserve semantic fidelity without introducing detectable artifacts [8]. ImF's Chain-of-Thought–aligned behavioral optimization is explicitly proposed for extension to multimodal LLMs, indicating its potential to embed ownership identifiers in cross-modal reasoning traces—e.g., joint visual-textual justification paths in VLAMMs or multimodal chain-of-thought in audio-visual question answering—thereby anchoring provenance in functional equivalence rather than static signal injection [13]. Layered Watermarking (LWM), instantiated by BanglaLorica, provides a template for cross-domain generalization beyond English: explicitly designed for non-Latin, morphologically rich languages (Bangla) without linguistic customization, its three-tiered architecture—sampling-based, lexical, and syntactic layers—can be generalized to code (AST-guided structural substitution), audio (prosody-aware phoneme replacement), and vision-language (scene-graph–aligned token reordering), ensuring multi-layered signal resilience against domain-specific adversarial transformations [6]. CASIS's core framework—context-aware lexical substitution guided by NLI-based semantic

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relatedness—is explicitly proposed for future extension to multilingual watermarking, confirming its direct applicability to cross-lingual generalization and, by extension, to cross-modal settings where semantic relatedness can be computed across modalities via contrastive vision-language models (e.g., CLIP embeddings) [12]. Explanation-as-Watermark (EaaW) demonstrates empirical cross-domain generalization: validated on image classification (ResNet-18/CIFAR-10/ImageNet) and text generation (GPT-2, BERT/wikitext/bookcorpus/ptb/lambada), it employs task-dependent metric functions—predicted probability of ground-truth class for CV, average prediction probability of target tokens for NLP—enabling seamless extension to multimodal settings requiring heterogeneous verification logic [27]. Holmes has been validated for image captioning tasks, confirming cross-domain generalization beyond classification; future work explicitly targets extension to multimodal and video understanding tasks, supporting its role in multimodal provenance frameworks [30]. MOVE's future work includes extension to non-vision domains (e.g., NLP, audio) using domain-appropriate external features—indicating cross-domain generalization potential beyond text, such as code watermarking or audio synthesis [29]. TMFA's future work explicitly identifies extension to sequential language models (BERT, GPT) as critical, requiring development of text-based probing inputs and synthetic input synthesis methods aligned with transformer latent spaces—thereby bridging behavioral fingerprinting from vision to language domains [20]. The Unicode homoglyph method's future work includes extending homoglyph sets to non-Latin scripts (e.g., Arabic, Cyrillic) for multilingual support, laying groundwork for cross-lingual and cross-modal typographic watermarking where script-agnostic confusability serves as a universal carrier [4]. PJTC-Watermark's extension to other languages requires replicating the perceptual experiment with native speakers of English, German, and Chinese to quantify paragraph juncture ambiguity ranges and validate cross-linguistic applicability of structural prosody as a watermark carrier—highlighting that prosodic invariance, like syntactic or semantic invariance, constitutes a modality-agnostic robustness anchor [35]. Finally, $\text{arxiv}2510_10493outlinesacomprehensivemultimodalandcross-domaingeneralizationpathway: extensiontomulti-languageandcross-languageattribution, dynamic/runtimefeatures(executiontraces), andrecursivetraining/distillationsettings^{-}conj domainresiliencemustencompassnotonlystaticoutputsbutalsotemporal, behavioral, andcompositionalprovenanceac domaingeneralizationproceedsfromarchitecturalprinciples^{-}notadhocadaptations^{-}enablingprovablyrobust, interop compliantprovenanceacrossthefullspectrumofLLM-generatedartifacts.$

8.3 Adaptive Threat Resilience and Provable Defenses

Adaptive threat resilience necessitates the co-design of watermarks and fingerprints with formally grounded, empirically validated defenses against LLM-powered adversaries—including self-information-guided paraphrasers (SIRA), gradient-based removal techniques (GRI-2), and ambiguity attacks that exploit statistical or structural dependencies across queries—where resilience is not merely empirical but certified through statistical guarantees, cryptographic hardness, or information-theoretic bounds. SAC exemplifies this co-design paradigm by constructing trigger sets via adversarial search over input space to elicit maximal output-distribution divergence, while its theoretical analysis demonstrates that correlation matrices over softmax outputs remain invariant under fine-tuning, pruning, transfer learning, and model extraction; this preservation of internal representational geometry enables certified robustness against adaptive adversaries who aim to erase behavioral signatures without collapsing functional utility—and future work explicitly proposes formal analysis of correlation preservation under model transformations as a pathway toward provable guarantees [38]. Holmes advances this direction by embedding provable statistical guarantees directly into the verification protocol: its paired T-test on prediction-difference vectors $(S - P)$ yields bounded false-positive rate $\alpha = 0.01$ and empirically measured error rates as low as 10^{-20} under Data Distill on ImageNet, establishing statistically grounded decision thresholds that resist LLM-powered distillers and ambiguity attacks reliant on distributional mimicry rather than exact memorization [30]. Dual-Verification (DVF) formalizes resilience against ambiguity attacks by requiring simultaneous satisfaction of Label Accuracy (LA) > th_{LA} and Bit Error Rate (BER) < th_{BER}, thereby enforcing joint success criteria that eliminate one-dimensional vulnerabilities exploited by MatchPrefix or MatchSubstring protocols; its gradient-guided GradCAM embedding creates computational intractability for reverse-engineering the copyright-bit-to-input mapping without access to the autoencoder generator's latent space—providing an empirical basis for certified resistance against inversion-based ambiguity attacks [31]. MetaV co-designs fingerprint construction and verification to resist obfuscation—including weight compression, fine-tuning, and distillation—but explicitly acknowledges the absence of provable defenses against gradient-based removal or ambiguity attacks, identifying certified robustness under such

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threats as a critical open gap demanding game-theoretic modeling of adversary-verifier interaction [40]. MOVE demonstrates adaptive threat resilience against saliency-based detection, STRIP, and trigger synthesis attacks by decoupling ownership verification from trigger-label associations and instead leveraging external-feature–induced prediction-difference vectors; its future work proposes formalizing theoretical bounds on external feature uniqueness and stealing resilience—aligning with provable defenses against adaptive adversaries who reconstruct behavioral signatures via gradient analysis or query-response profiling [29]. SIRA provides both empirical motivation and theoretical scaffolding for adaptive co-design: its attack success probability is bounded by lemmas on Shannon self-information perturbations under small-δ deviations, establishing a formal basis for designing self-information-aware watermark embedding—where signal strength is dynamically modulated to minimize detectable information leakage—and for differential privacy-inspired signal injection, wherein noise calibrated to token-level sensitivity ensures provable indistinguishability under adaptive query-based probing [21]. WALL's orthogonal basis function indexing and hash-driven vocabulary permutation enable certified robustness against LLM-powered paraphrasers and GRI-2: by decoupling signal location from lexical positions and embedding non-overlapping high-SNR signals via discrete cosine transform modes, it supports differential privacy-inspired watermarking where perturbation magnitude is calibrated to inference uncertainty rather than fixed thresholds [5]. ImF co-designs steganographic outputs and Chain-of-Thought–aligned inputs to resist Generation Revision Intervention (GRI), implementing a dual-phase defense—Security Review (input detection) and CoT Optimization Instruction (response redirection)—that preserves behavioral signatures under white-box gradient-based removal while improving TruthfulQA accuracy, thereby transforming functional utility into a resilience amplifier and establishing a model for provably secure behavioral watermarking under adaptive adversaries [13]. DAWN introduces a cryptographically grounded notion of unremovability: its HMAC-SHA256–based deterministic watermarking ensures that any attempt to suppress or erase the backdoor label via adversarial fine-tuning, gradient masking, or logit manipulation necessarily degrades surrogate model utility below usability thresholds, yielding certified robustness guarantees against Sybil attacks, collusion, and adaptive removal strategies targeting query-response consistency [10]. TMFA identifies critical gaps in behavior-based fingerprinting resilience, proposing fingerprint-erasure techniques during fine-tuning, batch normalization parameter hardening, and input distortion mechanisms to disrupt latent-space correlation exploitation—thereby enabling certified defenses against adversaries who reconstruct or filter behavioral signatures via gradient analysis [20]. Crucially, provable defenses require breaking cross-prompt consistency: Water-Probe demonstrates that static keys enable adversaries to infer model identity across queries, thus motivating context-aware adaptive keys—where watermark parameters are dynamically derived from prompt semantics or user-session state—to achieve provable LLM imperceptibility and eliminate black-box identification vectors [9]. Collectively, these advances point toward a new generation of adaptive, provably resilient mechanisms—not as isolated algorithms but as composable, game-theoretically informed primitives—where watermarking and fingerprinting are no longer passive signals but active, certified defenses co-evolving with the threat landscape.

8.4 Standardization, Evaluation, and Sociotechnical Infrastructure

The maturation of LLM watermarking and fingerprinting hinges on the co-development of technical standards, verifiable infrastructure, and regulatory-aligned evaluation frameworks—where interoperability, auditability, and evidentiary rigor are engineered into the architecture rather than retrofitted as compliance afterthoughts. ISO/IEC JTC 1/SC 42—the international standardization body for AI—has been explicitly identified as the critical venue for defining formal watermarking profiles that specify minimum detectability thresholds (e.g., AUC 0.95), fidelity tolerances (e.g., perplexity 0.5), robustness requirements (e.g., Attack Success Rate < 5% under SIRA v2), and verifiability constraints (e.g., black-box, training-free, keyless operation) necessary to treat watermark evidence as legally sufficient for copyright enforcement or EU AI Act Article 28 compliance . TokenizedModel directly advances this standardization mandate by implementing verifiable watermark logging APIs with on-chain verification records, decentralized ownership registries via Tokenized Model NFTs (embedding model hash and owner address), and immutable, timestamped blockchain entries that satisfy EU AI Act Article 28 audit-log requirements—thereby enabling regulator-admissible provenance without centralized control or proprietary infrastructure [14]. Similarly, TM-PoT motivates ISO/IEC JTC 1/SC 42 standardization of Proof-of-Training profiles through specifications for public test dataset sampling (D_{test}), trajectory serialization formats, verifiable synthetic data fidelity metrics (e.g., $\eta > 1$ threshold), and standardized audit-log interfaces compliant with EU AI Act Article 28—ensuring

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third-party verification without private data exposure while preserving cryptographic soundness and GDPR alignment [33]. PersonaMark supports SC 42–compliant deployment through its training-free, black-box–compatible verification and user-ID–bound structural hashing—enabling standardized watermark logging APIs where dependency-tree hashes serve as timestamped, immutable, and auditable evidence of user-attributed generation without requiring model access, secret keys, or internal state inspection [7]. In-Context Watermarking (ICW) further enables platform-controlled, consent-aware provenance logging through instruction-engineered prompts, requiring no model modification and supporting fine-grained attribution (document ID, timestamp, client credentials) with built-in audit logging and consent management—making it directly applicable to ICW-deployed systems subject to EU AI Act Article 28 compliance mandates [17]. MIM—the Model Interpolation and Merging benchmark—advances toward SC 42 alignment by proposing verifiable watermark logging protocols for compositional deployment scenarios, including standardized merging algorithms (Task Arithmetic, TIES-MERGING, DARE), fixed test sets (50 GSM8K, 50 StrongReject-small), and deterministic Verification Success Rate (VSR) computation—thereby enabling reproducible, infrastructure-grounded assessment of watermark survival in real-world hybrid architectures [39]. SIRA v2 serves as a candidate for standardized attack suite integration in community benchmarks (e.g., WaterBench), offering reproducible, model-agnostic, resource-efficient, and semantically faithful paraphrasing evaluation across languages and models—providing the adversarial foundation required for SC 42–compliant robustness certification [21]. To ensure cross-method comparability, standardized LLM-native evaluation protocols must include knowledge-graded threat tiers (Tier A: uninformed adversary; Tier B: method-aware; Tier C: code-aware), deterministic output parsing requirements, unified Java re-implementations of watermark algorithms, and synthetic testbeds preserving Unicode structure—including confusable character distributions and script-specific rendering invariants—to eliminate implementation-dependent variance and enable rigorous forensic comparison [34]. These initiatives collectively constitute a sociotechnical infrastructure stack: from ISO/IEC JTC 1/SC 42–profiled watermarking specifications, to verifiable logging APIs supporting both in-context and parameter-space embeddings, to decentralized ownership registries enabling regulator-admissible audit trails—all converging on a single objective: transforming watermarking and fingerprinting from experimental artifacts into interoperable, auditable, and legally defensible pillars of trustworthy LLM ecosystems.

8.5 Governance-Ready Technical Frameworks

Governance-ready technical frameworks operationalize regulatory compliance—particularly under the EU AI Act Article 28, NIST AI RMF assurance levels, and GDPR data minimization principles—by embedding verifiability, auditability, consent, and decentralization directly into the architecture of watermarking and fingerprinting systems. FedZKP constitutes a foundational governance-ready framework that implements federated zero-knowledge proofs (ZKPs) for watermark verification: it embeds the SHAKE-256 hash $h = H(A||y)$ of an xLPN problem instance (A, y) into BatchNorm scaling parameters via hinge-loss regularization, and enables ownership verification through succinct, trust-free ZKPs of the secret solution (s, e)—thereby providing on-chain-verifiable ownership certificates without exposing weights, training data, or cryptographic secrets [32]. Its design satisfies EU AI Act Article 28 requirements by generating immutable, timestamped, challenge-response–based audit logs with KB-scale communication and ms-level verification time, supporting decentralized ownership registries without centralized control or trusted third parties [32]. TokenizedModel extends this governance infrastructure through three complementary mechanisms: (i) Tokenized Model—a smart-contract–enabled on-chain ownership certificate that immutably links model hash, owner address, watermark type (feature-based or backdoor), and contribution metadata to Ethereum-compatible blockchains; (ii) Watermark-as-a-Service (WaaS) APIs featuring built-in audit logging, consent management via smart-contract initialization (e.g., user opt-in triggers on-chain registration), and stake-weighted voting for verification consensus among distributed stakeholders; and (iii) federated watermark customization enabling privacy-aware personalization across multi-client deployments while preserving cryptographic binding and regulatory traceability [14]. Critically, TokenizedModel's WaaS architecture supports both in-context and parameter-space embeddings, allowing closed-API providers to publish verifiable proofs while open-weight users retain full model access—thus bridging interoperability gaps between proprietary and community ecosystems [14]. FedSOV advances governance readiness by enabling federated zero-knowledge proofs as a future work direction, supporting on-chain ownership certificates via cryptographic binding of aggregated public keys to hash digest h, and delivering EU AI Act Article 28–compliant

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audit trails through verifiable, challenge-response–based ownership verification logs that require no model access beyond standard inference calls [28]. TM-PoT serves as a foundational provenance substrate for governance frameworks: its zero-knowledge verification and chain-based certification of training trajectory integrity directly enable FedZKP (via federated ZKPs of synthetic data fidelity) and Tokenized Model (via on-chain anchoring of trajectory hashes and D_{test}-aligned fidelity certificates), while its WaaS integration provides built-in PoT verification modules for real-time regulatory attestation [33]. The PoT architecture itself—certifying six quantitative properties of training dynamics (monotonic accuracy, GMM-constrained initialization, etc.) using only public checkpoints and auxiliary datasets—provides a tamper-resistant, data-private provenance signal that underpins all three frameworks, ensuring GDPR-compliant auditability without private data exposure [41]. Finally, arxiv2406$_1$1106$proposescryptographicwatermarkingadvances^{-}includingzero-knowledgeverifiablewatermarksandprivacy-awarepersonalizationframeworkssuchasfederatedwatermarkcustojurisdictionalenforceability, therebytransformingwatermarkingfromatechnicalartifactintoalegallydefensible, sintegratedlogging, andzero-knowledgeattestationsconvergetosupportaccountable, interoperable, andregulation-compliantLLMdeployment.$

9 9. Conclusion

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This survey reconfigures LLM watermarking and fingerprinting as interdependent security primitives grounded in formal distinctions, empirically validated trade-offs, and sociotechnical constraints—not isolated technical interventions. The three-axis taxonomy—*Target*, *Embedding Mechanism*, and *Verification Paradigm*—unifies fragmented conceptualizations by rigorously separating text-level attribution (e.g., SynthID-Text's tournament sampling [22], WALL's hash-permuted vocabulary [5], PersonaMark's dependency-structure hashing [7]) from model-integrity verification (e.g., PLMmark's contrastive weight perturbations [26], SAC's trigger-induced correlation matrices [2], ImF's Chain-of-Thought–aligned behavioral optimization [13]) while accommodating hybrid paradigms such as TokenizedModel's dual-locus embedding [14] and In-Context Watermarking's prompt-space anchoring [17]. These distinctions expose irreducible tensions: Easymark's Theorem 3.4 establishes a formal impossibility—erasure resistance under adversarial semantic distortion requires sacrificing either fidelity or detectability [8], while Water-Probe demonstrates that statistical distinguishability in watermark detection inherently compromises model anonymity under repeated black-box querying [9]. Empirical benchmarks confirm these limits—SynthID-Text achieves TPR@1%FPR > 0.95 monolingually yet collapses to AUROC 0.62 under English–Chinese RTT [22]; X-SIR improves CWRA robustness by +0.221 AUC but increases detector memory footprint and degrades compatibility with quantized inference engines [3]; PLMmark withstands fine-tuning yet fails catastrophically under model merging, whereas ImF remains merging-resilient but succumbs to white-box gradient-based removal (GRI) [13]. The standardized five-dimensional evaluation framework—operationalizing detectability (z-score confidence hierarchy [9], TPR@1%FPR [22]), fidelity (perplexity delta [7], ROUGE-L [22], human-rated fluency [22]), robustness (ASR under SIRA v2/GRI-Adapt/CWRA/MIM), computational overhead (latency delta [5], detector memory [18]), and deployability (API compatibility [5], language support [3], open-source status [22])—reveals systematic patterns across 32 techniques: high detectability consistently trades against cross-lingual resilience; parameter-space fingerprints resist fine-tuning but collapse under architectural fusion; behavior-based methods scale to open-weight deployment yet leak via trigger-set reconstruction. Forward progress hinges on provable foundations—not heuristic adaptation—such as information-theoretic models of watermark capacity under semantic distortion, multimodal generalization (VLA-Mark's vision-language alignment [25], PersonaMark's AST-guided code attribution [7], arxiv2510$_1$0493$'sJavaScriptwatermarking$[19]$), adaptivethreatresilience(SAC'scorrelation-invarianttriggers$[38], $DVF'sdual-verificationagainstambiguityattacks$[31]$), andgovernance-readyinfrastructure(FedZKP'szero-knowledgeproofs$[32], $TokenizedModel'son-chainownershipcertificates$[14], $PoT'szero-knowledgetrajectorycertification$[41]$).Suchadvancestransformwateronsintoco-designed, enforceablepillarsforauditabilityunderthe EU AI Actand NIST AI RMF, legalaccountabilityinAPI, open-weight, andfederateddeployments.$

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